Are the phases real? Distinguishing bounded interaction groups from spatially structured drift in central Mississippi Valley decorated ceramics

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Abstract

The phase, one of archaeology's foundational units, rests on the silent premise that assemblages more similar to nearby than to distant neighbors record a bounded society. While the validity of phases has been questioned, the rationale for assuming these units is rarely tested or evaluated. Neutral transmission produces the same clustering with no bounded groups since drift concentrates similar styles nearby. For the late pre-contact central Mississippi Valley, we test whether decorated ceramic similarity records reflect bounded interaction groups or spatially structured drift. We compare drift and bounded-groups processes on the region's geographic data, calibrated to the record's small samples and time-averaging using a signal-recovery experiment. In both the St. Francis basin and the southeast-Missouri assemblages, drift reproduces the structure without bounded groups. Across the wider valley, it generates far fewer coherent groups than the named phase scheme draws. We find that Parkin, often considered the center of a polity, is a transmission bridge rather than a capital. Tested directly against the other named phases, it shows at most a weak signal at the edge of the drift expectation. A sharper boundary appears only between the central and lower valleys, confounded with chronological turnover and analytical history. Before similarity is interpreted as social, a record must be shown to be able to distinguish groups from drift.

Keywords: cultural transmission, neutral models, frequency seriation, isolation by distance, Mississippian archaeology, Parkin phase

Resumen

La fase, una de las unidades fundamentales de la arqueología, descansa sobre una premisa tácita: que los conjuntos más similares a sus vecinos cercanos que a los distantes registran una sociedad delimitada. Aunque la validez de las fases ha sido cuestionada, la justificación para asumir estas unidades rara vez se pone a prueba o se evalúa. La transmisión cultural neutral produce el mismo agrupamiento sin grupos delimitados, ya que la deriva concentra estilos similares en áreas cercanas. Para el valle central del Misisipi de finales del período precontacto, evaluamos si la similitud de la cerámica decorada registra grupos de interacción delimitados o deriva estructurada espacialmente. Comparamos dos procesos generativos, deriva y grupos delimitados, sobre la geografía real de los conjuntos, calibrando la comparación a los tamaños de muestra pequeños y al promediado temporal del registro mediante un experimento de recuperación de señal. Tanto en la cuenca del St. Francis como en los conjuntos del sureste de Misuri, la estructura se reproduce mediante deriva y no requiere grupos delimitados, y en el conjunto del valle la deriva genera muchos menos grupos coherentes de los que distingue el esquema de fases nombradas. Encontramos que Parkin, a menudo considerada el centro de una entidad política, es un puente de transmisión más que una capital. Evaluada directamente frente a las demás fases nombradas, la fase Parkin muestra a lo sumo una señal débil en el límite de lo que predice la deriva. Una frontera más marcada aparece solo entre el valle central y el valle inferior, confundida con el recambio cronológico y la historia analítica. Antes de interpretar la similitud como social, debe demostrarse que un registro puede distinguir los grupos de la deriva.

Palabras clave: transmisión cultural, modelos neutrales, seriación de frecuencias, aislamiento por distancia, arqueología misisipiana, fase Parkin

Introduction

The late pre-contact record of the Lower Mississippi River Valley is traditionally divided into a series of spatially and temporally bounded units known as phases (Figure 1). These units include examples such as the Kent (House 1991), Nodena (Dan F. Morse 1990), Parchman (Starr 1984), Tipton (Smith 1990), Walls (Smith 1990), Jones Bayou (Smith 1990), and Parkin (Phyllis A. Morse 1990) phases. The Parkin phase stands out among these units because it includes the Parkin site, the largest of the moated, palisaded towns in the St. Francis basin of northeastern Arkansas. The chronicles of the Hernando de Soto expedition place it as Casqui, the seat of a paramount chief, which the expedition encountered in 1541 (Hudson et al. 1990; Mitchem 2001; Phyllis A. Morse 1990). On the strength of its size, fortification, platform mound, and ethnohistoric identification, Parkin is routinely interpreted as the center of a Mississippian polity, on the long-standing assumption that a dominant mound center marks the apex of a ranked society (Peebles and Kus 1977). This chiefdom is inferred to have drawn tribute from the smaller villages around it, an argument based on its position on agriculturally poor soil, which required surrounding communities to provide it with provisions (Phyllis A. Morse 1990). That interpretation is a specific claim about social structure in which the Parkin phase is treated as a bounded, hierarchically organized society with Parkin at its apex.

Whether such an entity existed at Parkin is an empirical question. Settlement morphology alone cannot answer this question, since the presence of a primary center of interaction, a palisade, and a monumental mound is equally compatible with a landscape of competing peer communities as with a centralized and consolidated hierarchy (King and Freer 1995; Rees 2001). The 16th-century historical chronicles describe Casqui and its neighbor Pacaha in open conflict, a relationship interpreted both as rivalry between competing peers and as a ranked order in which Casqui stood below Pacaha. Thus, if Parkin is indeed Casqui, the chronicles do not, by themselves, establish a single consolidated hierarchy over the Parkin phase (Hudson et al. 1990; Phyllis A. Morse 1990). We ask, therefore, what the archaeological record shows when it is measured rather than interpreted through an assumed polity model. The record shows no evidence of hierarchy. Within the basin, settlement size is graded rather than redistributive. The regional production of the capacity-linked chert hoes whose control would mark an economic apex was decentralized, and their exchange followed a down-the-line gradient rather than central redistribution (Cobb 2000). As we show below, the decorated-ceramic record also carries no clear signature of interaction closing in on bounded groups.

The more revealing question is why Parkin was taken for a polity in the first place. The answer lies in the unit through which the record is interpreted, a matter the renewed debate over archaeological systematics has brought back into focus (Lyman 2021). The Parkin phase is a culture-history construct, a set of assemblages grouped because their decorated-class frequencies seriate together and because the sites cluster in space (Phillips et al. 1951). A social interpretation, however, goes beyond the definition. In the Phillips-Ford-Griffin framework, the phase is the basic space-time-cultural unit, defined by distinctive traits within a locality or region over a brief span, with no social criterion in the definition itself (Phillips and Willey 1953; Willey and Phillips 1958). Willey and Phillips (1958:48–49) even set out the syllogism that would make a phase a society, component equals community, and phase equals a set of components, and then rejected it. In practice, however, though the chances are against archaeological phases having much social reality, they judged them as such and acted as if they did. Williams (1954), defining the southeast-Missouri phases on which the regional scheme rests, adopted the social equation, writing that a phase probably equals a society. That equation, though disavowed at the source, is typically repeated in regional practice. This assumption about the unit phase is its silent premise. It holds that assemblages more similar to one another than to their neighbors record a bounded, mutually interacting social group. In the case where one can identify a group with a primate center, the phase then becomes a polity. The polity interpretation is, therefore, built into the unit before any independent evidence of hierarchy is weighed.

The phase construct has been questioned on exactly this ground. Fox (1998) showed, for example, by cluster analysis and ordination, that the southeast-Missouri phases fail as both groups and classes, with their member assemblages assigned to statistical clusters at random rather than segregating by phase. Mainfort (2003) reached the same verdict by a different ordination, finding that the ordination of the late-period assemblages provides little support for the validity of the named phases, which overlap rather than forming discrete clusters. Lipo (2001) questions the nature of phases through frequency seriation, showing that the assemblages are structured rather than panmictic, and arguing that the resulting spatial groups have no warrant as societies. He (2001:60) suggests "the phases themselves might be simply arbitrary spatial and temporal slices of a single interacting population that is much larger in size."

The silent premise, however, is tested only rarely and overlooks a strong alternative that offers a potentially falsifiable explanation for the observed patterns in the archaeological record. Viewed through the lens of social

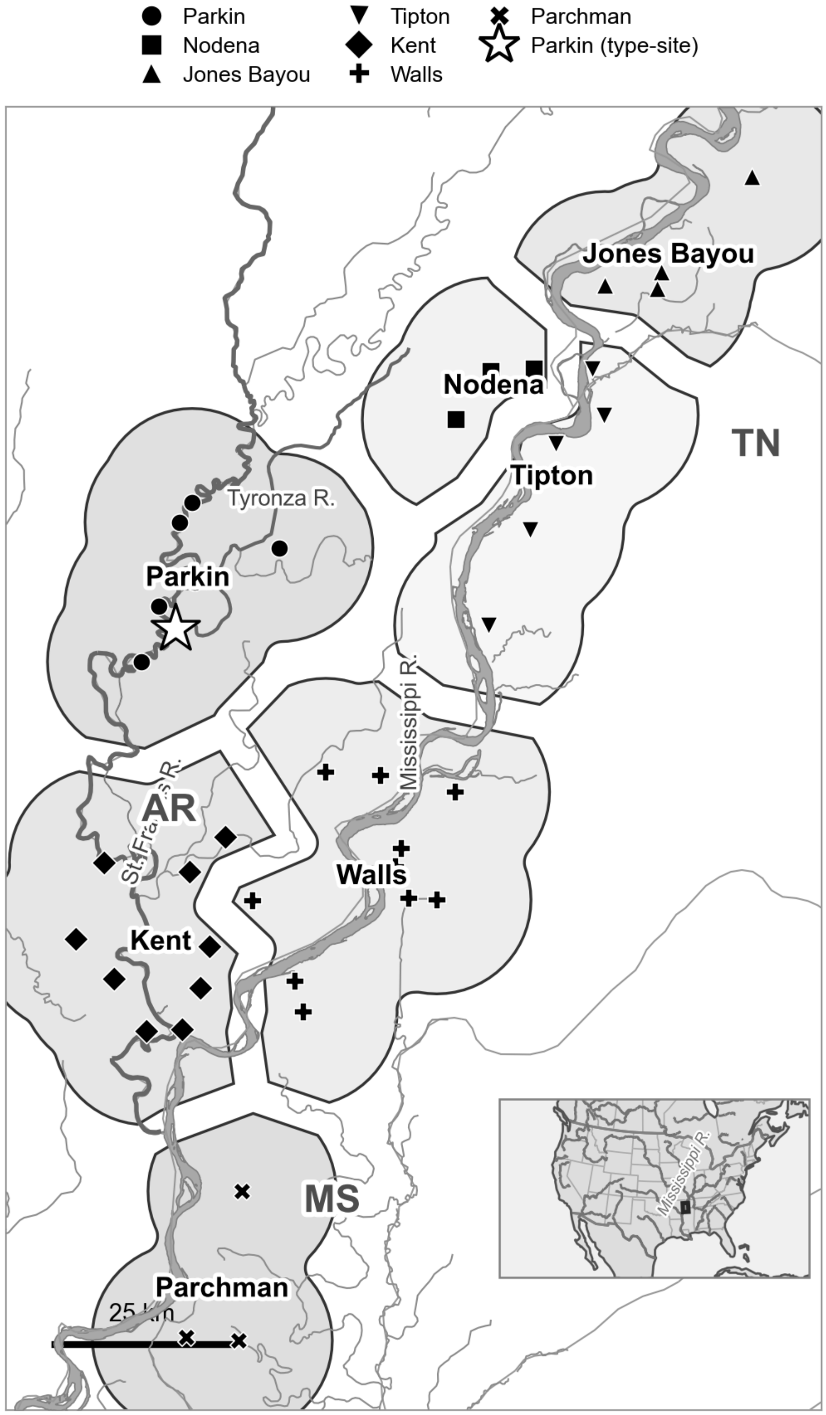

Figure 1. Decorated-ceramic assemblages of the central Mississippi Valley and the culture-historical phase scheme this paper tests. The seven Late Mississippian phases (Parkin, Nodena, Jones Bayou, Tipton, Kent, Walls, and Parchman) are shown as discrete territories, a Voronoi partition of the assemblages Mainfort (1996b) assigns to each phase, dissolved by phase and drawn with a gap between adjacent territories for legibility, over a local hydrology base (the St. Francis, Tyronza, and Mississippi rivers) with state and county boundaries. Each assemblage is plotted with a marker corresponding to its phase, and Parkin is marked with a star. The bounded territories illustrate the premise the paper tests: that such areas host distinct, internally interacting societies. The transmission analysis finds instead that drift structured by geography explains the impression of groupness. The mapped assemblages fall in Arkansas, Tennessee, and Mississippi; Missouri lies just north of the frame. Inset: location in North America.

learning and models of cultural transmission, interaction is always continuous yet highly variable. Interaction is structured by distance and geography, as well as by social relationships. The task is to show how apparent differences between groups, such as phases, can arise under continuous interaction rather than requiring bounded groups. Under the neutral model of cultural transmission, decorative style behaves as a selectively neutral trait whose frequency drifts in a finite population of interacting learners (Bentley et al. 2004; Dunnell 1978; Eerkens and Lipo 2007; Fogarty et al. 2024; Neiman 1995). When those learners interact more with near neighbors than with distant ones, drift on the spatially structured population generates similarity that is highest at short distances and fades with distance. Viewed at larger scales and divided into discrete units, that gradient looks like difference and clustering, with no bounded groups anywhere (Lipo et al. 1997, 2021). Based on this account, a phase can be the ordinary residue of continuous, distance-structured interaction. The very pattern the culture-history tradition interprets as a society, assemblages near each other in space and in style, is what spatially structured drift predicts. Telling a real interaction group apart from spatially structured drift requires a test that the tradition never applied, because the seriation coherence and spatial proximity that define the phase are equally the products of drift (Riede et al. 2019).

Decorated ceramics provide an instrument for the test. Because stylistic variants are copied among potters through social learning networks, the spatial and temporal structure of decorated-class frequencies reflects past interactions. A genuine social boundary, had one existed, would make learning appear conformist within groups and divergent between them, fragmenting the single drifting pool that continuous interaction maintains (Lipo et al. 1997, 2010, 2015; Neiman 1995). The late pre-contact central Mississippi Valley is a favorable place to look, both because its decorated assemblages are well documented and because European contact and the subsequent demographic collapse ended the sequence before any later reorganization could overwrite it (Cobb and Butler 2002). We first focus on the St. Francis basin, the setting of Parkin (Figure 1), and then extend the test to other late-period phases of the valley.

The argument rests on a choice about what the ceramic record is. We do not treat an assemblage as a degraded snapshot of a past behavioral moment, to be restored by correcting for the processes that formed it (Schiffer 1972). We treat it as what it is, a time-averaged accumulation whose decorated-class frequencies are the population-level residue of social learning, to be measured and explained directly rather than reconstructed (Lipo et al. 2015; Neiman 1995). This is an evolutionary position in which the record is the evolving phenotype to be explained rather than a window onto a vanished behavioral system (Dunnell 1978, 1980, 1995a; Ladefoged and Graves 2000), and it converges with the recognition that the deposit is not a frozen tableau (Binford 1981). The record's coarse temporal resolution is not a defect to be removed but the scale at which long-term, population-level processes become legible (Bailey 2007; Behrensmeyer et al. 2000; Holdaway and Wandsnider 2008; Perreault 2019; Stern 1994). Time-averaging, therefore, sets the resolution of the question rather than disqualifying it. A reorganization of transmission large enough to matter over the two centuries of the Parkin phase should register in the averaged frequency structure. Thus, a drift-versus-groups test must measure that structure directly rather than an imagined moment behind it.

No single transmission signature can settle the question, because each is consistent with several processes. A departure from neutral copying, a spatial cline, or a rise in between-group variance can arise from drift, conformity, or environmental patchiness alike. This is an example of equifinality, in which the same pattern can be reached by different processes, undermining one-signature inferences (Barrett 2019; Crema et al. 2016; Kandler and Shennan 2015; Mesoudi 2021). Therefore, we compare processes in a generative manner by simulating neutral spatial drift and bounded groups on the real coordinate layout of the assemblages. Using simulated data, we ask which measures can reproduce the observed structure, using four transmission signatures: departure from neutral copying, breakdown of frequency-seriation coherence, between-group variance, and spatial assortativity. Because archaeological assemblages are time-averaged and sampled, our measurements may potentially reflect differences that are artifacts of sampling rather than of transmission. Consequently, we calibrate our comparisons to the record using a signal-recovery experiment in which synthetic assemblages are matched to the real number, sample sizes, and time-averaging. We then inject a known signal of tunable strength and measure how strong a real boundary would have to be to register. This calibration enables us to conclude that the presence or absence of group structure is real, not a failure of resolution. This calibration is the transferable contribution, because before any similarity structure is interpreted as social, a record must be shown to be able to distinguish drift from groups at all.

The argument proceeds in three steps. First, we evaluate the Parkin as part of a hierarchy and find that no line of evidence, settlement, exchange, or transmission supports it. Second, we ask what generates the St. Francis basin phase structure, and show that neutral drift on the basin's geography reproduces the observed signatures. A bounded-groups process would leave a far sharper structure than the data demonstrate. Hence, we conclude that the phase is a portion of a continuous interaction structure rather than a closed group with a boundary, and Parkin is a transmission bridge rather than the core of a group. Third, we widen the test to the other late-period phases of the valley and ask whether decorated-ceramic boundaries exist. A boundary that exceeds neutral drift appears only between the central-valley and lower-valley blocks, and even there, it is confounded with the early-to-late turnover of the class repertoire. It cannot yet be interpreted as a synchronic social division. At the scale where the phase is drawn, it is largely indistinguishable from spatially structured drift. We conclude that Parkin is best understood as a prominent node in a continuous field of interaction rather than the capital of a polity.

Background

The phase, the silent premise, and the drift alternative

In evaluating the phase concept, we take a materialist approach to cultural transmission. This approach frames the question of phases differently from how the central-valley record is usually interpreted. Archaeologists are traditionally trained to assume the past was made of bounded units: the phase, the polity, the society. These units are treated as discrete entities with members and edges. A cluster of similar assemblages is considered one of them. In contrast, we treat transmission instead as a continuous, variable field of copying events among potters, with no intrinsic boundaries. Interaction falls off with distance, and learning is more local in some places and times than in others. A group, on this view, is not a thing with a boundary but a region where copying is locally intensified. Similarity is a matter of degree, set by how copying is distributed across space. The patterns of interaction can also be closed by degree. In this scenario, interaction narrows until potters copy mostly within a subset and rarely outside it, producing something that resembles the bounded groups that archaeologists assume.

The phase gives the central valley record its social structure. Operationally, a phase is a set of assemblages that seriate together and cluster in space. It is interpreted, however, as a social entity, a bounded, interacting community, on the premise that assemblages that are more alike within a phase than across its edges are generated by individuals who interacted mostly within (McNutt 1996; Morse and Morse 1983; Phillips et al. 1951; Williams 1954). The premise is not an unreasonable possibility. Under the neutral transmission model, it would hold if learning were confined within social boundaries, so that decoration drifted in separate pools and diverged between them. The difficulty is that the same pattern arises with no boundary at all. When interaction declines with distance, spatially structured drift concentrates similarity among near neighbors. This is the isolation-by-distance expectation for cultural similarity (Lipo et al. 2021; Neiman 1995; Shennan et al. 2015). A cluster of similar assemblages, therefore, need not record a bounded group. What would mark a real bounded group is closure beyond what distance predicts, and that excess is the signal we must test for. Our operational question is whether the structure of similarity records bounds interaction or spatially structured drift. We can treat decoration transmitted preferentially within a group and divergent between groups as a marker, rather than a neutral style drifting in a shared pool. A marker is the ceramic proxy for interaction closing into a bounded group, so the drift-versus-groups question, then, is effectively a question of neutral style versus marker.

Why ceramics, and the neutral-vs-marker distinction

Measures of variability in decorated ceramics can capture the structure of social interaction because stylistic variants are copied among potters through social learning networks. Under the neutral model of cultural transmission, stylistic variants behave like selectively neutral alleles whose frequencies are governed by innovation and drift in a finite population of interacting learners (Bentley et al. 2004; Bentley 2008; Dunnell 1978; Eerkens and Lipo 2005; Neiman 1995). Neutral style is the null condition: assemblages drawn from a single interacting community share one drifting pool, and their decorated-class frequencies admit a single battleship-curve seriation (Dunnell 1995b; Lipo 2001; Lipo et al. 2015; Lipo and Madsen 2001; O'Brien and

Leonard 2001). Assortment, the tendency for like to learn from like, so that interaction sorts into distinct groups, changes this. If learning becomes restricted within emerging social boundaries, decoration shifts from the expectations of neutral, stylistic variation toward a constrained set of possibilities, transmitted by conformist copying within groups and divergent between them, and the single transmission field fragments (Crema et al. 2016; Henrich 2001; Henrich and Boyd 1998). When this process plays out, the restricted pool of variation can come to look like "markers" of bounded social groups, even if no sharp bounds existed.

The obstacle is that no single frequency pattern identifies its generating process. A departure from neutrality can reflect conformity, an anti-conformist novelty bias, a content bias, or simply the fact that durable objects, rather than people, are the units that accumulate in a deposit (Crema et al. 2016, 2024; Henrich 2001). The problem is more complex in archaeological assemblages, which are time-averaged, because aggregation over the use-life of a deposit inflates richness, flattens evenness, and biases neutrality statistics by an amount that depends on the mean lifetime of the variants (Madsen 2012; Premo 2014). The result is equifinality: multiple transmission processes produce the same population-level pattern, so a single statistic cannot recover the process that generated it. A discriminating signal, where it survives, lies in the rare-variant tail of the frequency distribution rather than in its common classes (Carrignon et al. 2019; Kandler et al. 2017; Mesoudi 2021; O'Dwyer and Kandler 2017).

The central Mississippi Valley case and two competing hypotheses

The late pre-contact central Mississippi Valley supports two interpretations of the same record, and they make opposite predictions for the transmission signature. The first holds that the region was moving toward greater complexity. In northeastern Arkansas, the Parkin phase (ca. 1400 to 1600 CE, with a thin protohistoric tail is consistent with the basin radiocarbon reported below) is a dense cluster of fortified, moated St. Francis-type towns along the St. Francis and Tyronza rivers. It is dominated by the Parkin site, a roughly 17-acre palisaded center whose platform mound rises more than 6 m, which the De Soto chronicles place as the seat of the paramount of Casqui (Dye and Cox 1990; Mitchem 2001; Phyllis A. Morse 1990). Based on this information, Parkin's location on agriculturally poor soil suggests that it drew tribute from smaller villages, a hallmark of an emerging chiefly hierarchy (Phyllis A. Morse 1990). Because contact and the demographic collapse associated with the protohistoric "Vacant Quarter" (Williams 1990) interrupted the sequence, any such transition would have been truncated before it could consolidate and overwrite its own beginnings (Cobb and Butler 2002; Morse and Morse 1996). We treat this truncation as a working premise rather than a settled fact, since the timing and extent of the Vacant Quarter abandonment relative to the Parkin phase remain debated (Cobb et al. 2024; Cobb and Butler 2002). The second interpretation holds that the central valley was a stable, non-consolidated landscape of competing peer polities that never reorganized into a regional hierarchy (Anderson 1996; Beck 2003; Blitz 1999; King and Freer 1995; Lulewicz 2019; Milner 1998; Rees 2001). The chronicles describe Casqui and its neighbor, Pacaha, in open conflict, which fits within a landscape of competing peers. The same sources have also been interpreted as ranking Casqui below Pacaha, and the chronicle ethnography is thin and filtered (Hudson et al. 1990; Mitchem 2001; Phyllis A. Morse 1990). The two possibilities for Parkin cannot be separated by settlement morphology alone. They can, however, be separated by determining whether the transmission record shows interaction closing into bounded groups, which is the test we now construct.

Distinguishing true group structure from spatially structured drift

The test rests on one principle: a process that genuinely reorganizes a population into groups leaves a mark on every aspect of how pottery styles were copied, whereas a process that merely mimics group structure marks only some aspects. The reorganization at issue, which we call the closure of interaction, is central to the rest of the paper, so we define it plainly. In an open community, potters can learn a design from anyone they encounter, near or far, so styles circulate through one connected population. Interaction closes when that circulation narrows: potters come to copy mostly from others inside their own group and rarely from anyone outside it, until a once-continuous field of social learning breaks into separate pools that drift apart. Closure is the process a genuine bounded interaction group undergoes, and we treat its strength as a quantity that runs from none (one fully open field of learning), through partial, to complete (sealed, non-interacting groups). That strength is what we set out to test and to measure. In the synthetic experiments below, we treat it as a parameter, *s*, from 0 (pure drift, no closure) to 1 (strongly bounded, sharply separated groups). We can evaluate four potential signatures, each a different consequence of closure, and we state each in plain terms before naming their formal measures.

The first asks whether potters copied decorated styles at random or with a bias. Under unbiased copying, where a potter is as likely to reproduce any design she encounters as any other, two independent ways of gauging an assemblage's diversity agree. One is based on homozygosity, a measure borrowed from genetics. This property reflects the probability that two sherds drawn at random from an assemblage share the same decorated class, which is high when a few classes dominate. The other is based on richness, the number of classes in which sherds are identified. Each yields an estimate of the same underlying transmission parameter, and under random copying, the two estimates coincide. Conformity, in which potters disproportionately copy whatever is already common, is a process that will concentrate frequency to a few ceramic classes. It raises homozygosity, lowering the homozygosity-based estimate while leaving richness little changed, pulling the two apart. A taste for novelty does the reverse. The disagreement between the two estimates, expressed as their ratio, therefore both detect a departure from random copying and show which way it leans (Neiman 1995; Shennan and Wilkinson 2001). This matters here because a bounded group that enforced a shared decorative marker would copy conformist behavior even to the edges of its boundaries. As a result, the two estimates would diverge, whereas a single pool of styles drifting at random would keep them equal.

The second asks whether the assemblages all belong to one interacting community or to several, and it is the most direct test of fragmentation. It rests on frequency seriation, the classic archaeological method of ordering assemblages so that each decorated class's relative frequency rises to a single peak and then falls away, the shape long known as a battleship curve. That single-peaked rise and fall is what a style traces when it is introduced, becomes popular, and is abandoned within one community, drawing on a shared pool of designs. In classical uses of seriation, all available assemblages were typically pooled into a single seriation, and deviations from the frequency guidelines were dismissed as sampling error (Dunnell 1970). Sampling error certainly needs to be accounted for (Lipo 2001). Classical practice, however, ignores the most valuable aspect of the method: a *failure to seriate together* indicates that locales are far enough "out of step" with the surrounding flow of interaction and social learning that their interaction history has interesting structure (Lipo et al. 1997, 2015).

A set of assemblages is "co-seriable" when a single ordering can make every class follow the same single-peaked shape simultaneously. When all assemblages arise from a single continuously interacting population, such an ordering exists. When learning splits into separate groups, each with its own repertoire and its own independent rise and fall of styles, no single ordering can satisfy every class at once, and the assemblages break into several mutually incompatible orderings. The deterministic frequency-seriation procedure of Lipo, Madsen, and Dunnell (2015) finds the complete set of these maximal co-seriable groups algorithmically, without the analyst arranging the assemblages by eye (Lipo et al. 1997). The number of groups and the number of groups any one site joins directly measure the structure of past interaction. A single large group means one connected field of learning, a proliferation of small groups means a fragmented one, and a site that falls into several incompatible orderings is a bridge linking otherwise separate sequences rather than the closed core of a bounded community.

The third asks how much of the variation in decoration lies between groups rather than within them. The measure is a cultural version of Wright's F_ST (Wright 1951), a standard population-genetic index of group differentiation. In cultural transmission, between-group variation in socially learned traits routinely exceeds that of their genetic counterparts (Bell et al. 2009). We compute it here using the Gini-Simpson diversity measure, the chance that two sherds drawn at random from an assemblage are of different classes. F_ST is the share of the total diversity that is due to differences between groups rather than to variation within them. When every assemblage draws from one shared, drifting pool of styles, between-group differences are small, and F_ST sits near zero. As groups come to favor distinct repertoires, the between-group share grows, and F_ST climbs (Neiman 1995). It serves here as the most direct measure of how far decoration has sorted itself by group, with one important caveat developed below. Stochastic drift on a spatially structured population generates between-group variance on its own. Subpopulations retain different levels of diversity depending on their size and connectivity. Small or sparsely connected areas drift toward depleted, idiosyncratic repertoires, while larger, well-connected ones remain richer. The resulting differences appear as distinct interacting groups, even though they are produced solely by spatial position (Lipo et al. 2021). The measure must therefore be calibrated against that spatial-drift null rather than taken at face value.

The fourth asks whether socially close assemblages are more alike than assemblages that are merely nearby on the map. Similarity normally fades with distance, because potters interact more with near neighbors than with distant ones, a gradient known as isolation by distance (Neiman 1995; Shennan et al. 2015; Wright 1943). A genuine social boundary adds a step to that gradient. Under this condition, pairs of assemblages within the same cluster are more similar than pairs in neighboring clusters, even when the two pairs span the same geographic distance. We capture that step as a boundary-excess statistic, the gap between within-cluster and between-cluster similarity once distance is held constant, which a plain correlation of similarity with distance cannot isolate. It is useful here because this measure separates a real social edge from the ordinary spatial fade, again with the caveat that drift on a finite network can itself produce a positive excess (Lipo et al. 2021).

Each plausible alternative to a bounded interaction group involves only a subset of these signatures (Table 1). Environmental patchiness, in which similar people occupy distinct ecological zones, produces spatial structure with no departure from neutral transmission and no loss of seriation coherence. Aggregated signaling, the bottom-up regime in which a conformist bias flattens diversity within a single community, produces a departure from neutrality without boundary formation, so seriation remains coherent and between-group variance remains low. Smooth, deterministic drift with distance-limited interaction produces an isolation-by-distance cline (Shennan et al. 2015; Wright 1943). This pattern appears as a gradual fading of similarity with distance, as people interact mostly with near neighbors, without the sharp boundaries or departures from neutrality that mark restricted, assortative learning. A bounded interaction group, in which one assortative process restricts transmission along every dimension, is the only process that drives all four signatures together at full strength. But the table is idealized in a way that matters here: two of the four signatures, between-group variance and the distance-controlled boundary excess, are also driven by stochastic neutral drift on a spatially structured population, which generates patchy divergence and a positive boundary excess with no bounded group anywhere (Results, Figure S5). The linked movement of these two measures cannot separate a real interaction group from drift. Therefore, we cannot depend on these signatures alone. Instead, we must compare the two processes generatively by simulating each one, neutral spatial drift and bounded groups alike, using the real geography of the deposits. Using the data from the simulations, we can then ask which reproduces the observed structure, treating the signatures as diagnostic of where each process leaves its mark.

Table 1. What each process does to the four transmission signatures, and which signature this record can measure. A filled circle (●) marks a signature, while an open circle (◦) is one where it is not measurable. The process columns give the idealized discrimination logic: under smooth, deterministic processes, only a bounded interaction group drives all four, and each alternative drives a characteristic subset, the pattern an idealized simulation confirms (Figure S4). Two features of the table show why that logic is not the test we run on the record. The smooth-drift and stochastic-drift columns separate textbook isolation-by-distance, which drives none of the four, from realistic neutral drift on a finite network, which drives cultural F_ST and the boundary excess on its own, with no bounded group present (Figure S5). Stochastic drift thus reproduces the same two-signature pattern as environmental patchiness, so a positive result on either signature cannot by itself separate a bounded group from drift or patchiness. The final column reports the recovery experiment introduced below. In this way, the four signatures are therefore candidate diagnostics screened against the record, and the surviving discrimination is carried by cultural F_ST, assessed against the drift null, and by the generative drift-versus-groups comparison, not by convergence of the table's rows.

Signature	Bounded group	Environmental patchiness	Aggregated signaling	Drift (smooth)	Drift (stochastic)	Detectable on this record?
Departure from neutral transmission	●	◦	●	◦	◦	No

Signature	Bounded group	Environmental patchiness	Aggregated signaling	Drift (smooth)	Drift (stochastic)	Detectable on this record?
Seriation-coherence breakdown	●	○	○	○	○	No
Between-vs within-group variance (cultural F_ST)	●	●	○	○	●	Yes
Spatial assortativity (boundary excess)	●	●	○	○	●	No

The two drift columns make the difficulty that we face explicit. Smooth, deterministic isolation-by-distance produces only a pattern of decay in similarity with distance. In addition, the distance-controlled boundary excess is near zero under it, so it does not drive any of the four signatures. Stochastic neutral drift on a finite network is less forgiving. It generates patchy divergence that raises cultural F_ST and registers as a positive boundary signal even in the absence of a social boundary, a mimicry that the record-matched controls demonstrate (Results, Figure S5). Aggregated signaling, by contrast, raises the neutrality departure within a single undivided pool without forming the between-group boundaries that the other signatures require.

If closure were to occur, it would accumulate across the sequence. On idealized data, genuine convergence should take the form of a trajectory: a coherent, same-direction shift in all four signatures across the ordinal sequence of assemblages, concentrated where and when interaction closes into bounded groups. The logic is only as strong as the independence of the signatures, because four measures that were merely redundant converge trivially. As a consequence, we chose to validate the test using simulated assemblages before applying it. This validation seeks to confirm that convergence occurs only when interactions cluster into a bounded group, and not under any mimic. We also use the simulation to assess whether the four signatures are tightly correlated under such closure, yet nearly independent under the mimics. In this way, their joint movement should be informative rather than automatic. The use of convergence logic has an archaeological precedent. Neiman (1995), for example, showed that two independently derived, theory-grounded measures of the same ceramic data, within-assemblage diversity and between-assemblage distance, converge on the same inference about past interaction. We generalize his approach and apply it to our four measures. Whether all four signatures, or only some, can actually do so on a record like ours is a separate question that we settle by calibrating the tests directly to synthetic data matched to the Parkin assemblages.

That calibration is a signal-recovery experiment (described in the Materials and Methods section, with full details in the Supplemental Text). We build synthetic assemblages matched to the real ones in number, sample size, and time-averaging, and inject a known group-closure signal of controlled strength. We then ask which signatures recover it and how strong the closure must be to register. This process is necessary for two reasons specific to this record: it has only 29 time-averaged assemblages, and assemblage size trends along the seriation axis, so a size-sensitive measure can move through sampling alone. Reading each signature against this record-matched standard, rather than against the idealized expectation, is what enables us to treat an absence of structure as real rather than as a failure of resolution. The procedure and its parameters are detailed in the Materials and Methods and the Supplemental Text, and Figure 4 reports the outcome.

Below, we describe the procedures, their methods, and outcomes. Figure 2 previews what the test distinguishes. It shows the sharp structure that a bounded group would leave against the diffuse pattern that drift produces, and why, on a record of this size and time-averaging, a weak within-basin boundary cannot be excluded.

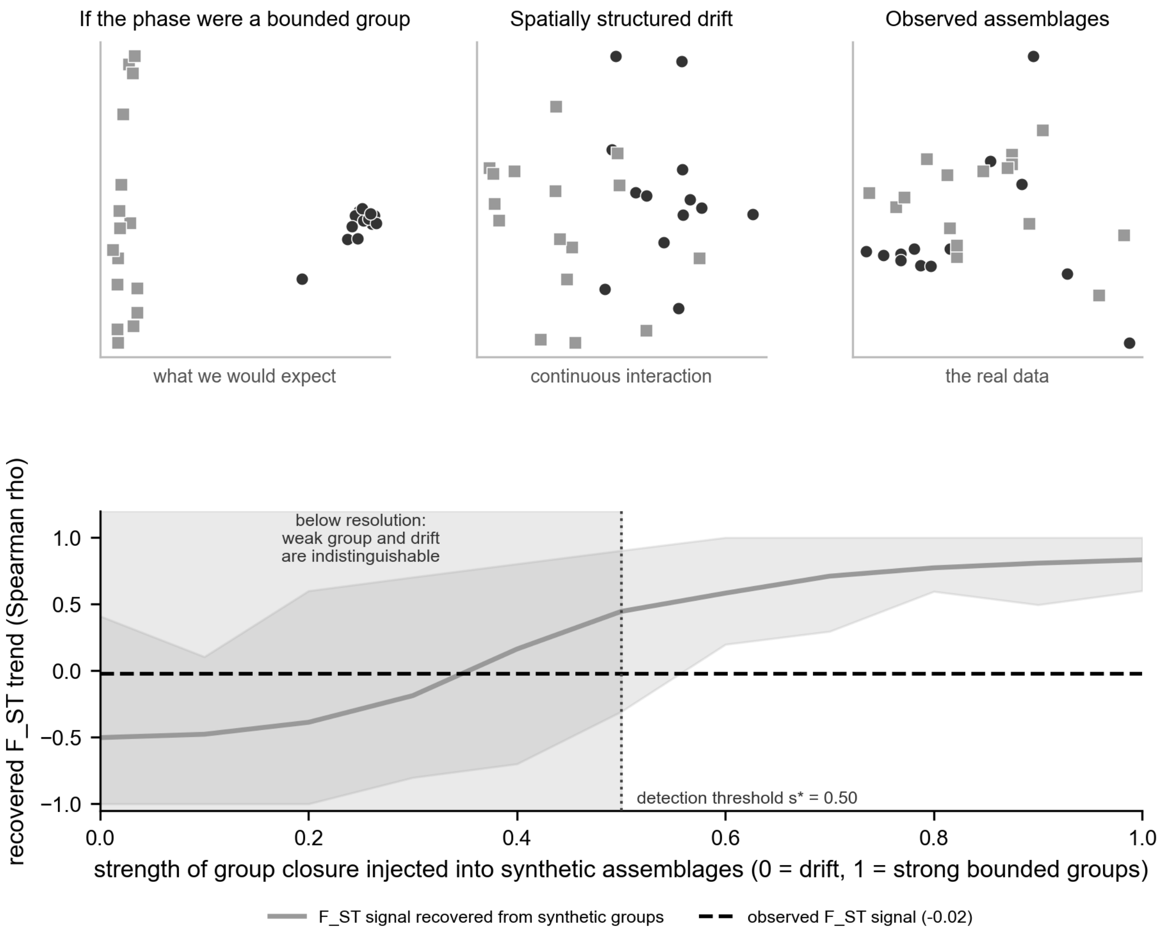

Figure 2. What the test distinguishes, and why the within-basin determination is hard, from synthetic assemblages matched to the record. Top: the basin assemblages in a multidimensional scaling of decorated dissimilarity (points colored by the k=2 spatial split) as they would appear if the Parkin phase were a bounded interaction group (left, sharp clusters), under spatially structured neutral drift (center, a continuous gradient), and as observed (right, a diffuse scatter resembling drift). Bottom: the cultural F_ST signal recovered from synthetic assemblages as the strength of injected group closure rises from 0 (drift) to 1 (strong groups), with the s=0 null band shaded behind it. The signal clears the detection threshold only at s* = 0.5. The observed value (dashed line) lies in the shaded region below s*, where a weak group and pure drift are indistinguishable at this record's size and time-averaging. This is why a weak within-basin boundary cannot be excluded.

A signature is trustworthy to the degree that it meets three conditions, usually treated apart. It should be theoretically sufficient. In this case, a cultural transmission model provides its assumptions, resulting in a theoretical claim rather than an analogy. Deterministic frequency seriation is the model case, because the unimodality and continuity it requires are exactly what the neutral model predicts of a single interacting community. Neiman's (1995) diversity and distance estimators are likewise derived from the same cultural transmission model (Lipo et al. 2015). It should be dynamically sufficient in Lewontin's (1974) sense, with its state variables and their transformation laws sufficient to carry the system from one state to the next. And it should be empirically sufficient. This condition means it is matched to the quality of the archaeological record and defined at the time-averaged resolution preserved by the deposit, rather than at a snapshot or generational resolution (Holdaway and Wandsnider 2008; Perreault 2019).

Thus, our four signatures are theoretically and dynamically served by the transmission model. Empirical sufficiency, whether a signature can actually register a signal at the resolution the deposit preserves, is not guaranteed by this framing, requiring us to test it rather than assume it. As we will describe below, at this record's 29 time-averaged assemblages, only cultural F_ST registers the injected signal, and even then, it must be compared against the stochastic-drift null rather than zero. Seriation-coherence breakdown, which only a bounded group drives in principle, is the cleanest discriminator on paper yet undetectable here. Therefore, measured as states along the sequence, the signatures are only partially dynamically sufficient, a gap we close in the discussion with tests that model the transformation between states.

Materials and methods

Data

The ceramic data that are the basis of the original phase definitions come from the decorated-class counts compiled by Phillips, Ford, and Griffin (1951) for the Lower Mississippi Valley Survey. We have compiled these data into an assemblage-by-class matrix of 266 surface assemblages (Lipo 2001). Each decorated class is a culture history type, a unit of measurement that the analysts defined to capture stylistic variation, not a native category assumed in advance. For our study, we restrict the analysis to the St. Francis basin of northeastern Arkansas, the setting of Parkin, defined hydrologically as the assemblages within 20 km of the St. Francis, Tyronza, and L'Anguille drainage (Figure 1). This yields 29 decorated assemblages for the transmission analysis and 92 sites for the settlement comparison. The basin is a geographic unit, not a single culture-historical phase: under the standard regional scheme, its assemblages fall into the Parkin phase together with sites assigned to the Nodena, Kent, Walls, and Parchman phases (Mainfort 2003; Phillips 1970). The test, therefore, asks whether interaction closed into bounded groups anywhere in the basin, across the phase divisions that culture history draws within it, and within the Parkin phase. The set excludes the lower Yazoo Basin to the south. A stricter watershed rule, keeping only assemblages nearer the St. Francis system than the Mississippi, gives 19. The verdict reported below is unchanged across both definitions. The two southern St. Francis-type sites the drainage rule excludes, Old Town and Blanchard, lie 28 and 80 km from the St. Francis system and belong to the Mississippi-proximal and lower Yazoo settings rather than to the St. Francis drainage. We georeferenced each assemblage based on maps from the Lower Mississippi Survey site files. We used settlement attributes, site size, mound count and height, and the presence of defensive ditches and St-Francis-type fortification, from a regional geographic database (Lipo and Dunnell 2007).

Two properties of the data bound what any transmission analysis can claim, and we treat them as constraints rather than nuisances. The assemblages are time-averaged surface assemblages, so they consist of a sample of an aggregate of pottery formed over the use-life of its deposit. Such aggregation distorts diversity and neutrality statistics in proportion to the mean lifetime of the variants (Madsen 2012; Miller-Atkins and Premo 2018). The neutral model's expectations are derived for a population of people copying one another. An assemblage, however, counts durable objects, the pots and sherds themselves, not the potters who made them. Because the record is object-mediated rather than person-mediated, the standard neutral expectation does not apply exactly and must be understood with that caveat in mind (Crema et al. 2024).

The availability of an absolute chronology is a requirement, though the existing radiocarbon corpus is richer than a bare five-date anchor. The Mainfort (2001) compilation yields 109 determinations for the central valley, 41 of them from seven St. Francis basin Parkin-phase sites and 19 from Parkin itself (House 1993; Mainfort 1996a; Sullivan et al. 2024). Calibrated against IntCal20 (Reimer et al. 2020), the basin determinations place the occupation in the fourteenth to sixteenth centuries (summed-probability median AD 1422, Parkin AD 1483) and show a decline across the contact interval, as detailed in the Supplemental Text. A calibration plateau across that interval still defeats year-scale resolution (Holland-Lulewicz and Ritchison 2021), so we order the assemblages by seriation rather than by calendar age, using the first axis of a correspondence analysis (Baxter 1994; Peeples and Schachner 2012; Smith and Neiman 2007). This ordination arranges assemblages along axes that summarize their decorated-class-frequency differences, as a relative chronological ordinate oriented so that higher positions correspond to later time (Figure 3). Pooling the dates of the five proveniences that match curated assemblages, the median calibrated ages correlate with seriation position at

Spearman +0.70 (n = 5, 27 dates pooled), an improvement on the +0.50 of the single-date anchors. Four of the five sites fall in calendrical order, with one late-dated site departing from it. The axis, therefore, tracks calendar time imperfectly. The trajectory we test, therefore, is an ordinal measure, not a calendar rate. The discriminating quantity, the size-controlled F__ST trend, is a magnitude invariant to the axis orientation, so only the directional descriptions of the other signatures depend on the anchor. We return to both limits in the discussion.

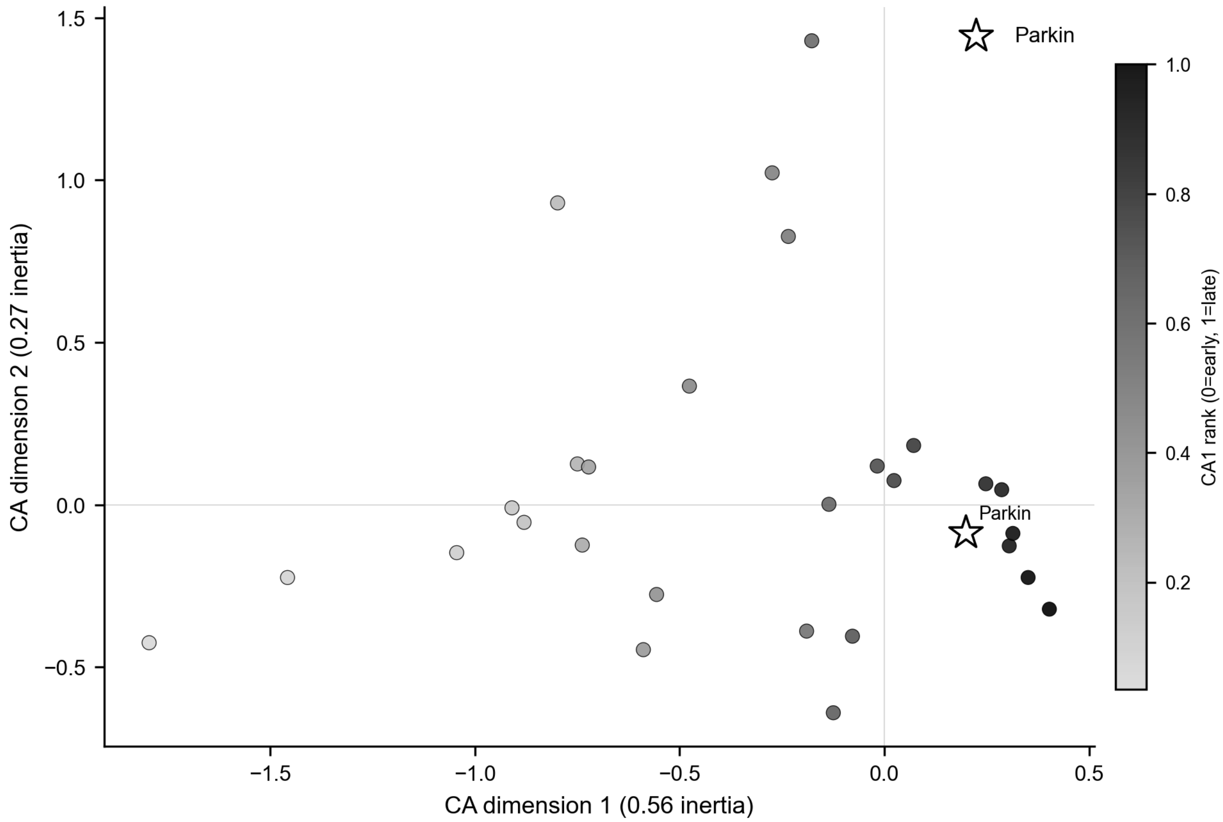

Figure 3. Correspondence-analysis seriation of the 29 basin decorated assemblages, plotted on the first two CA dimensions of the decorated-class frequencies and colored by CA1 rank (the relative chronological ordinate, oriented by the five available radiocarbon dates so that higher rank is later). Parkin is starred. The assemblages form a gradational continuum rather than discrete clusters, echoing the independent NMDS result of Mainfort (2003).

The four signatures

We computed each signature with a published or standard estimator and combined them only at the final step. Departure from neutral transmission follows Neiman's (1995) application of the infinite-alleles model to ceramic assemblages. We estimated the transmission parameter in two ways: from the unbiased homozygosity of an assemblage and from its number of classes, using the Ewens sampling formula (1972). This approach enables us to estimate the number of classes expected under random copying and to take the ratio of the two estimates as the signature. Under neutral drift, the two estimators agree. Conformist transmission lowers the homozygosity-based estimate relative to the richness-based one, and anti-conformist novelty bias does the reverse, so the ratio both detects a departure and indicates its direction (Shennan and Wilkinson 2001). Because pooling assemblages from divergent groups cancels the signal, we computed departures within spatial clusters and averaged them.

Seriation coherence follows an iterative deterministic frequency-seriation method of Lipo, Madsen, and Dunnell (2015). A set of assemblages is co-seriable if some ordering renders every decorated class both unimodal and continuous, and the method returns the complete set of maximal co-seriable groups. Because checking every possible ordering becomes computationally intractable beyond about fourteen assemblages, IDSS builds solutions agglomeratively: it starts with the valid three-assemblage seriations and extends each by adding assemblages that keep every class unimodal and continuous, until no further additions are possible (Lipo et al. 2015). Each class's departure from a single peak, or a break in its continuity, is assessed against bootstrap confidence intervals for the class frequencies, so it counts as a real break only when it exceeds the sampling error. Small, sparsely sampled assemblages have wider intervals and therefore admit more orderings. A single group denotes a single interacting community. A proliferation of small, non-co-seriable groups indicates a fragmented one. We report the number and size of these groups and, for each assemblage, the number of distinct groups it joins, which identifies assemblages that bridge otherwise incompatible orderings.

The third signature, the variance partition, is a cultural analog of Wright's F_ST, computed on Gini-Simpson diversity, the chance that two sherds drawn at random from an assemblage are of different classes (Bell et al. 2009). It is the difference between the diversity of the pooled assemblages and the mean within-group diversity, divided by the pooled diversity, taken across spatial clusters. It rises from zero as groups come to hold distinct decorated repertoires.

The fourth signature, spatial assortativity, measures whether ceramic similarity follows social rather than geographic distance. For each pair of assemblages, we computed the Brainerd-Robinson similarity (Brainerd 1951; Robinson 1951), a standard index ranging from 0 to 200 that quantifies how similar two assemblages are in their class proportions. We tested its association with inter-site distance using a Mantel (1967) permutation test, which asks whether two distance matrices, here ceramic dissimilarity and geographic distance, are related. A Mantel correlation cannot separate a sharp social boundary from a smooth distance decay. As a result, we also computed a boundary-excess statistic, which is the gap between mean within-cluster and between-cluster similarity after controlling for distance. Smooth isolation by distance returns a boundary excess near zero, while a genuine social boundary returns a large positive value.

Two corrections are essential before any of these signatures is interpreted. Assemblage sample size trends with seriation position, and several estimators are size-sensitive, so we subsample every assemblage to a common count (i.e., rarefaction) (Hurlbert 1971) before scoring each signature as its rank correlation with its ordinal position. Which signatures can then be trusted at the record's resolution is not assumed but established by the recovery experiment described next, in which a known closure signal is injected into the real configuration.

Criterion validation by simulation

The argument uses four simulations, each answering a different question. An idealized validation tests whether the four signatures, in principle, separate a bounded interaction group from its three mimics on large, clean synthetic assemblages (Figure S4). A record-matched recovery test assesses which signatures can detect a known, injected closure at the real record's sample sizes and time-averaging (Figure 4). A positive control test that generates neutral spatial drift or bounded groups reproduces the structure observed in the basin (Figure S5). A forward demonstration runs neutral drift on the real site geography with no boundary imposed, counts how many phase-like groups it produces, and checks whether the named phases exceed that number (Figures 8 and 9). The first two establish what the test can resolve on a record of this size; the second two identify which process the data match. We describe the validation and recovery here and the two drift simulations in the next subsection.

We validated the test in two stages. The first establishes the principle. A generative model copies decorated variants under neutral transmission, the random-copying process in which learners reproduce variants in proportion to their current frequency (Bentley et al. 2004; Neiman 1995), with three tunable controls: the between-group divergence of class profiles, the strength of within-group conformity, and a spatial rule placing group profiles on the landscape. Four generators instantiate the four idealized processes of Table 1, the bounded group and its three mimics (Mesoudi 2021). Run blind through the four-signature procedure across twenty random seeds, only a bounded interaction group drives all four signatures together, and the signatures are tightly correlated under closure yet nearly independent under the mimics, so their joint movement is informative rather than an artifact of redundancy (Supplemental Text, Figure S4). That test uses large, clean assemblages, and it shows what the test can do in principle, not what it can do on a record like ours.

The recovery test is generative: we built synthetic pottery records in which we controlled whether group structure was emerging, dialed from none to strong, and then asked whether our four measures could see it. The synthetic records were matched to the real ones: the same number of assemblages, the same range of sample sizes, and the same time-averaging imposed by surface assemblages (Madsen 2012; Miller-Atkins and Premo 2018; Premo 2014). If chiefdom structure had really been emerging at Parkin, this experiment would tell us whether methods like ours, applied to data like ours, would have caught it. Concretely, we generated synthetic assemblages, one per real assemblage, at the real per-assemblage sample sizes, spatial clusters, and ordinal bins. Into them we injected a known group-closure signal of tunable strength s, with between-group divergence and within-group conformity rising together along the sequence to a maximum of s. We then asked which signatures recover it, a record-matched recovery design in the spirit of simulation-based inference calibrated to the data at hand (Crema et al. 2014; Kandler and Shennan 2015). Two properties of the record make this necessary. Assemblage sample size trends with seriation position (Spearman rho = +0.59), and several signatures are size-sensitive so that a size-dependent estimator can rise or fall along the axis through sampling alone. We therefore subsample every assemblage, synthetic and real, to a common count before scoring, thereby removing the confound. And the record is time-averaged, which we reproduce by blending adjacent-bin production within a window. The recovery experiment then asks, at the resolution and sample sizes this record preserves, which signatures can be trusted and how strong a closure would have to be to register. One aspect of the calibration matters here. Because subsampling the larger, later assemblages to a common count slightly inflates within-cluster diversity and so depresses F_ST toward later positions, the null with no closure (pure drift) itself carries a mild negative trend rather than a flat zero. The test is calibrated against that null distribution, not against zero, so the false-positive rate is held at five percent regardless, and the empirical value is compared against the same null.

The drift-versus-groups comparison

The central test compares two generative processes on the real geography of the assemblages. The neutral spatial-drift model follows the finite-population network model of Lipo, DiNapoli, Madsen, and Hunt (2021). Each assemblage is a node holding a finite population of learners that copies decorated-class variants from within the node and, at a low rate, from other nodes with a probability that decays with geographic distance. Innovation enters at a fixed rate, and deposition is pooled over a trailing window to reproduce the record's time-averaging. The bounded-groups model is identical except that interaction is concentrated within a small number of spatially contiguous groups, with only a low leak of copying between them, so that learning is restricted within boundaries. Both are run on the observed coordinate layout and subsampled to the observed per-assemblage sherd totals. Both are scored by the same four signatures, along with network modularity, a measure of how strongly the assemblages divide into separate clusters, and the distance decay of similarity. We swept the drift model's interaction range, between-node mixing rate, and innovation rate, and report the full sweep (Supplemental Text) rather than a single setting. To extend the test to the wider valley, we added the southeast-Missouri survey assemblages of Williams (1954), digitized from the survey tables. We georeferenced both the Phillips-Ford-Griffin and the Williams assemblages using their Lower Mississippi Survey grid designations against the digitized survey point files, so that the two regions share a single coordinate frame. Decorated classes were harmonized between the two typologies, both rooted in the Phillips-Ford-Griffin framework, into a common vocabulary, and the between-region comparison was run both on all decorated classes and on the classes shared by the two typologies (Supplemental Text).

The same neutral-drift model is also run forward to produce the phase-like groups of Figures 8 and 9. We run it forward in time on the real site geography with no boundary imposed, so any structure in the output comes from geography and time alone. Each site holds a finite population of learners. In each generation, a learner copies the decorated class of another learner, drawn from its own site with high probability and, at a low rate, from another site with a probability that decays with along-river distance. At a low rate, the learner instead adopts a newly invented class, so the repertoire turns over through time, and the assemblages can be seriated. Because the assemblages are not contemporaneous, we sample the simulated field time-transgressively. Each synthetic assemblage is drawn at the time-slice that matches its position on the seriation axis, accumulated over a trailing window to reproduce the record's time-averaging, then subsampled to the observed sherd count. We detect groups in the results using modularity on the Brainerd-Robinson similarity graph, following the same procedure applied to the real assemblages. Because the group labels differ between runs, we summarize Parkin's group as a probability: across 500 runs, the fraction of runs in which each assemblage falls into the same drift-detected group as Parkin. We report the number of groups, the between-group cultural F_ST, and the co-membership against the named phases, with the full parameter grid in the Supplemental Text.

External cross-check

The transmission test stands on its own, but a polity, or any closure of interaction into bounded groups, should also leave non-ceramic traces. We assembled three further lines against which to check it: one genuinely independent of the ceramic data (the lithic exchange economy), one that shares its collection substrate (settlement organization), and one that re-examines the same assemblages by a different method (an independent ordination). We assessed settlement organization using the rank-size distribution of mound area within the basin and the distribution of mound height, asking whether a single center dominates as a redistributive apex, the classic archaeological correlate of a ranked society, or whether size is graded (Peebles and Kus 1977). For the political economy of exchange, we drew on Cobb's (2000) regional analysis of the production and distribution of Mill Creek and Dover chert hoes, the capacity-linked industrial goods whose control would mark economic centralization, rather than measuring it ourselves. And we compared our seriation result to Mainfort's (2003) independent nonmetric multidimensional scaling of overlapping late-period assemblages, a published ordination of 39 assemblages from the central valley and adjacent areas that tests whether the named phases form discrete, bounded clusters or a gradational continuum.

Results

The four signatures distinguish bounded groups from drift in principle, but at this record's resolution, only one of them registers the difference. The idealized validation behaves as designed. When interaction is made to close into bounded groups, all four signatures move together, while drift and the other mimics each move only a characteristic subset. The four are tightly correlated under closure (mean absolute pairwise correlation 0.88) yet nearly independent under the mimics (0.19 to 0.29), so their joint movement is informative rather than redundant (Figure S4). The record-matched recovery is the more informative of the two (Figure 4). When a known group-closure signal is injected at the real configuration and scored after size control, only cultural F_ST tracks it monotonically, climbing from the null toward a strong positive trend as the injected strength rises. The injected strength s ranges from 0 (pure drift) to 1 (strongly bounded, sharply bounded groups). The neutral departure is non-monotonic in conformity, the spatial boundary is unresponsive at the three spatial clusters the data support, and seriation coherence responds only weakly. With the false-positive rate held at five percent, F_ST recovers a closure of strength s greater than or equal to 0.5 with power above 0.8, the threshold rising to 0.7 under the record's time-averaging. The other three signatures are not reliable discriminators at this resolution, and we treat them as weak corroboration only. At this record size and with time-averaging, the one signature that can distinguish a bounded group from drift is between-group variance.

Applied to the St. Francis basin, that one reliable signature shows no closure into bounded groups, and the apparent signals in the others are confounded by sample size (Figure 5). The raw trajectory along the seriation axis seems to dissociate. Two of its trends, however, are sampling artifacts: because assemblage size trends with seriation position, the size-sensitive estimators move with it. Once all assemblages are trimmed to a common sherd count via subsampling, cultural F_ST falls from a raw value of +0.60 to +0.01. Its uncertainty range runs from no closure to a moderate signal, so it is uninformative rather than a resolved positive. The neutrality departure is likewise size-sensitive (its raw -0.94 only weakens to -0.83 under size control). The recovery shows that it does not track closure in either direction here, so it, too, cannot be interpreted as a transmission signal. On the recovery curve, the size-controlled empirical F_ST corresponds to a nominal injected strength of about 0.35, below the resolution limit. The record, therefore, shows no detectable closure of moderate or greater strength. The negative is informative for a closure of moderate or greater strength, which would have registered and is absent. It cannot exclude a weaker closure that, given a record of this size and time-averaging, is underpowered to resolve. The positive control reported next

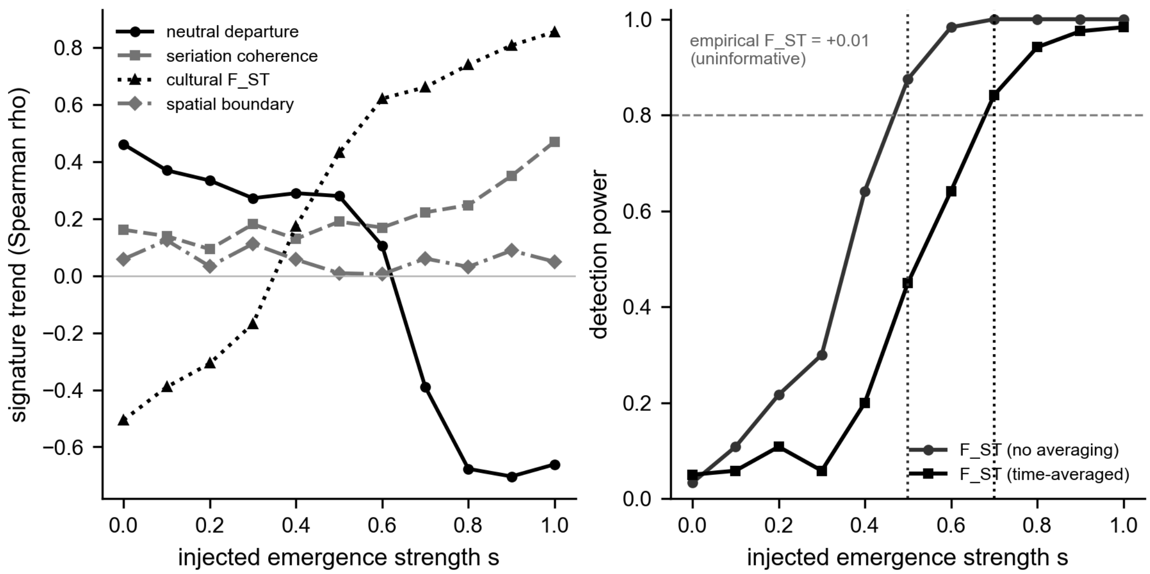

Figure 4. Record-matched, size-controlled signal recovery. Left: each signature's response (Spearman trend versus ordinal position) to a known group-closure signal injected at the real configuration of 29 assemblages and subsampled to a common count. Only cultural F_ST tracks the injected signal monotonically. The neutral departure is non-monotonic in conformity, the spatial boundary is unresponsive at three spatial clusters, and seriation coherence is weak. Right: detection power of the calibrated F_ST test against injected strength, with and without the record's time-averaging (false-positive rate 0.05). F_ST reliably recovers a closure of strength s greater than or equal to 0.5 (0.7 time-averaged). The size-controlled empirical F_ST (+0.01) falls below the resolution limit, its bootstrap interval spanning no closure to a moderate signal.

shows that the structure actually present is what neutral drift produces. The result is therefore not only that no bounded group is detectable, but that the structure present is what continuous, spatially structured interaction would produce.

Three further checks, all pointing the same way, add detail. The first follows each decorated class's rise and fall through the sequence and asks whether the changes resemble the undirected fluctuation of unbiased copying or the directed change a deliberate marker would exhibit. Seven of the nine testable classes are consistent with unbiased drift (the frequency increment test) (Feder et al. 2014). The second compares how the assemblages' overall diversity changes over time against the range that plain drift would produce, and finds that it stays within that range (a non-equilibrium neutral test) (Kandler and Shennan 2013). The third asks whether the assemblages grow more different from one another toward contact, as would fragmenting groups. They do not. Neiman's interassemblage distance shows no divergence trend along the sequence (Spearman rho = +0.37, not significant). The typical difference between spatial clusters is barely larger than the difference within them (a ratio of 1.13). The most diverse assemblages are also the most central, the pattern drift produces in a single shared community (Figure S1) (Neiman 1995). The verdict that no bounded-group closure is detectable also holds across the analyst's choices: across twenty-seven combinations of the basin's spatial cutoff, the number of ordinal bins, and the number of spatial clusters, no closure appears in any of them, and it holds equally under the watershed and corridor basin definitions. Consistent with the recovery experiment, none of these checks shows a resolvable departure from neutral transmission, and we treat them as corroboration of the size-controlled F_ST result rather than as independent signals.

A positive control identifies which of the two processes the basin's structure matches. Rather than rest on a failure to detect, we asked which of the neutral spatial drift and bounded groups reproduces the observed decorated-ceramic structure. Following the finite-population network-drift model of Lipo, DiNapoli, Madsen, and Hunt (2021), we simulated neutral copying on the actual coordinate layout of the basin assemblages, with interaction decaying over distance and the record's time-averaging imposed. We compared it against a

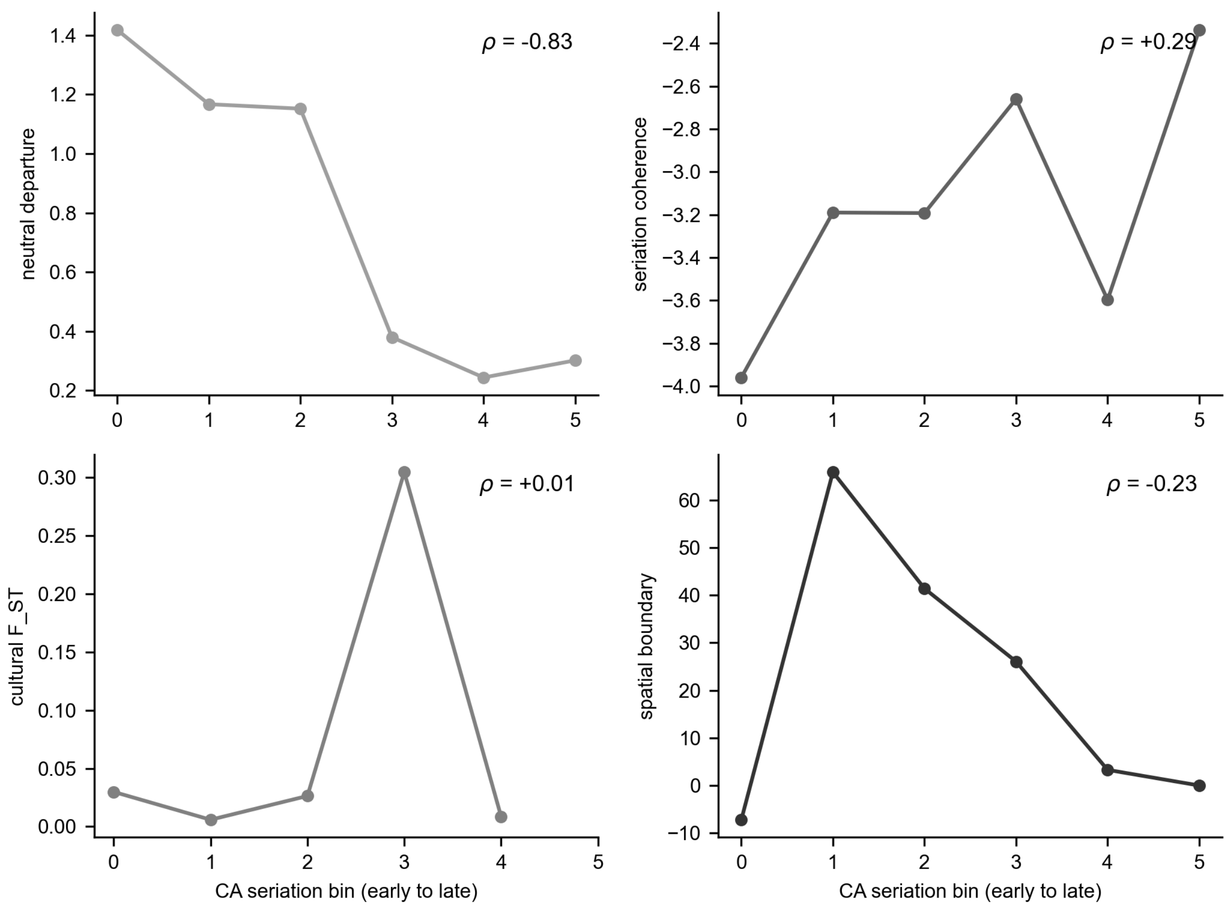

Figure 5. The empirical four-signature trajectory along the CA seriation axis for the St. Francis basin (29 assemblages, six bins), size-controlled by subsampling to a common count. The reliable signature, cultural F_ST, shows no trend (+0.01, down from a raw +0.60 that was a sample-size artifact). The neutral departure, spatial boundary, and seriation coherence are not reliable discriminators at this resolution (Figure 4) and are shown for completeness. Per-panel rank correlations are annotated.

bounded-groups process on the same geography. By sweeping the interaction range, the between-node mixing rate, and the innovation rate, neutral drift reproduces the observed distance decay, network modularity, cultural F_ST, and distance-controlled boundary excess. The region of parameter space that brackets all four at once is a sub-basin interaction range of about 24 km with low between-node mixing. We report the full sweep rather than a single tuned point: at shorter ranges, drift overproduces distance decay, and at higher mixing, it underproduces the structure, so the all-four match holds only in a plausible but specific corner of the space (Supplemental Text). The bounded-groups process, by contrast, overshoots the three structural signatures by roughly two to seven times across its range, predicting far sharper community structure than the data carry. The basin's decorated-ceramic structure is therefore reproducible by neutral spatial drift under plausible parameters. It does not require bounded groups, which would have left a sharper mark than is present (Figure S5). The control also bounds the spatial signature itself. A positive distance-controlled boundary excess, which we introduced to separate a sharp social boundary from smooth isolation by distance, is reproduced by stochastic neutral drift on a finite network, so on its own it does not certify a boundary. The same conclusion is reached for the phase construct from outside the transmission framework. Fox's (1998) cluster analysis and ordination of the southeast-Missouri assemblages that define the regional phases found them failing as both groups and classes, with assemblages assigned to statistical clusters at random rather than segregating by phase.

What it means for assemblages to be co-seriable is central to this signature, and the deterministic method is stricter than the seriations most archaeologists have in mind. In conventional practice, an analyst can almost always arrange a set of assemblages into an order in which class frequencies more or less rise and fall, treating the places where a class breaks that pattern as noise to be tolerated by eye. Deterministic frequency seriation removes that tolerance. It accepts an ordering only when every decorated class is at once unimodal, rising to a single peak and then falling away, and continuous, changing without an abrupt jump. It treats any break in a class's rise and fall as disqualifying that ordering unless the break is small enough to be explained by sampling error in the counts (Dunnell 1970; Lipo et al. 2015). Where a conventional seriation would accept one rough ordering and move on, the deterministic method asks whether even one ordering can satisfy every class at once, and reports when none can. A single community drawing on a shared, drifting pool of designs meets that requirement under a single ordering. When learning is partitioned among groups, each with its own rise and fall of styles, no single ordering can hold every class unimodal at once, and the assemblages resolve into several mutually incompatible orderings. The departures from a conventional seriation would be dismissed as minor irregularities, but, interpreted this way, they would reveal the signature of more than one interaction community. The number of maximal co-seriable groups, together with the number of groups any one assemblage joins, measures that fragmentation directly (Lipo et al. 1997).

Parkin is a seriation bridge, not the anchor of a bounded community. Deterministic seriation of the basin assemblages returns a deeply fragmented, bridge-dominated structure rather than a single coherent ordering. At a continuity threshold of 0.1, it yields 35 maximal co-seriable groups with a maximum size of four, and 17 of the 29 assemblages belong to more than one group (Figure 6). The absolute counts scale with the threshold, but the fragmented, overlapping topology is stable across thresholds. Parkin is one of these bridge assemblages, a member of four groups and ranked eighth of 29, positioned across orderings rather than at the center of one. The empirical pattern is the one Lipo, Madsen, and Dunnell (2015) reported, with Parkin at the intersection of multiple non-co-seriable orderings rather than at the center of a single closed interaction community. They interpreted that bridging position as a possible sign of incipient hierarchy. We interpret it differently: a site that bridges several orderings is a hub of interaction, not the seat of a bounded group.

Settlement does single Parkin out, but on the comparable evidence, its prominence sits within a graded distribution rather than standing apart as a redistributive apex. Parkin has the basin's tallest platform mound, at 23 feet, ranking first among the 55 basin sites with a recorded mound height (Figure 7). The distribution is graded, however, with no gap separating Parkin from the next centers, the tallest-to-second ratio being 1.0. The regional database under-records the moated, multi-mound type-site, coding its mound area and defensive ditch as zero (Mitchem 2001; Phyllis A. Morse 1990), so we base the settlement comparison on recorded mound height rather than on mound area. Parkin's roughly 17-acre site area would make it the largest center in the basin. That figure, however, is the total site area rather than the basal mound area on which the other sites are measured, so we treat it as indicative only. The settlement record thus shows a

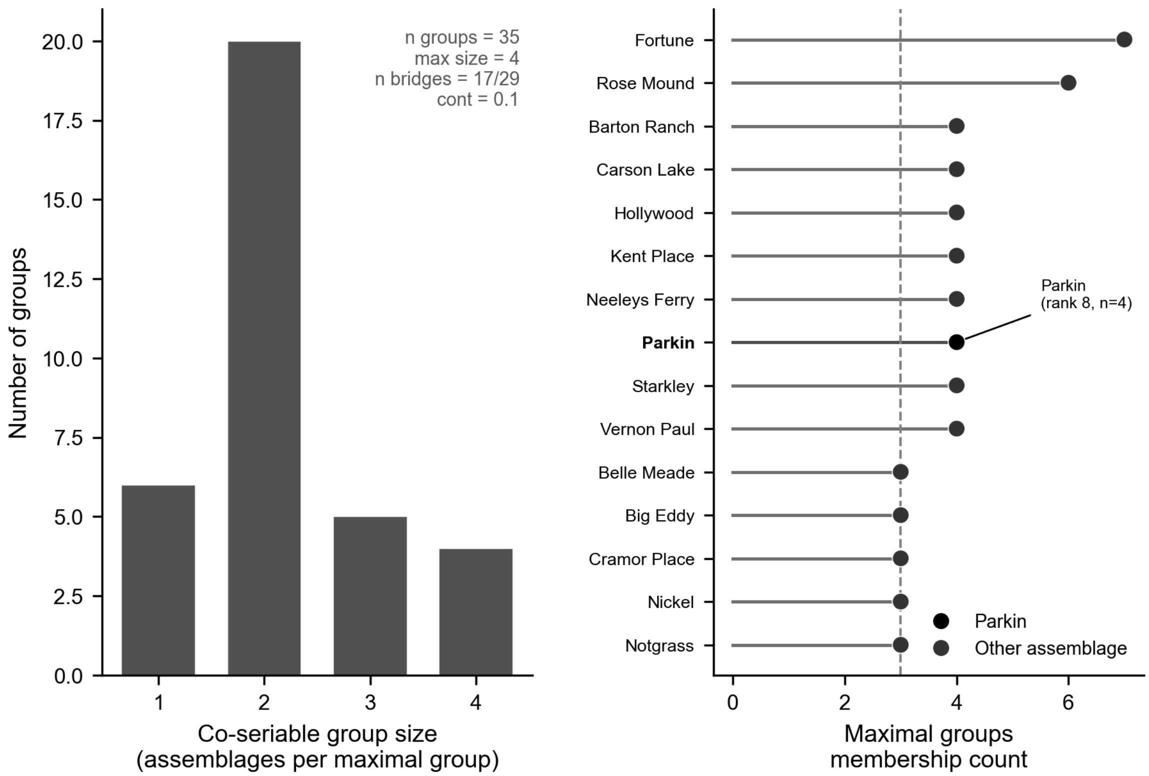

Figure 6. Deterministic frequency-seriation (IDSS) structure of the basin assemblages. Left: distribution of co-seriable group sizes, a deeply fragmented structure (35 maximal groups, maximum size four). Right: the most connected assemblages by group-membership count, with Parkin highlighted as a bridge (member of 4 groups, ranked eighth of 29), positioned across orderings rather than at the center of one bounded community.

prominent center within a graded size distribution, not an administrative tier set above the rest (Hally 1993, 1996), the same picture the transmission test returns (Figure 7).

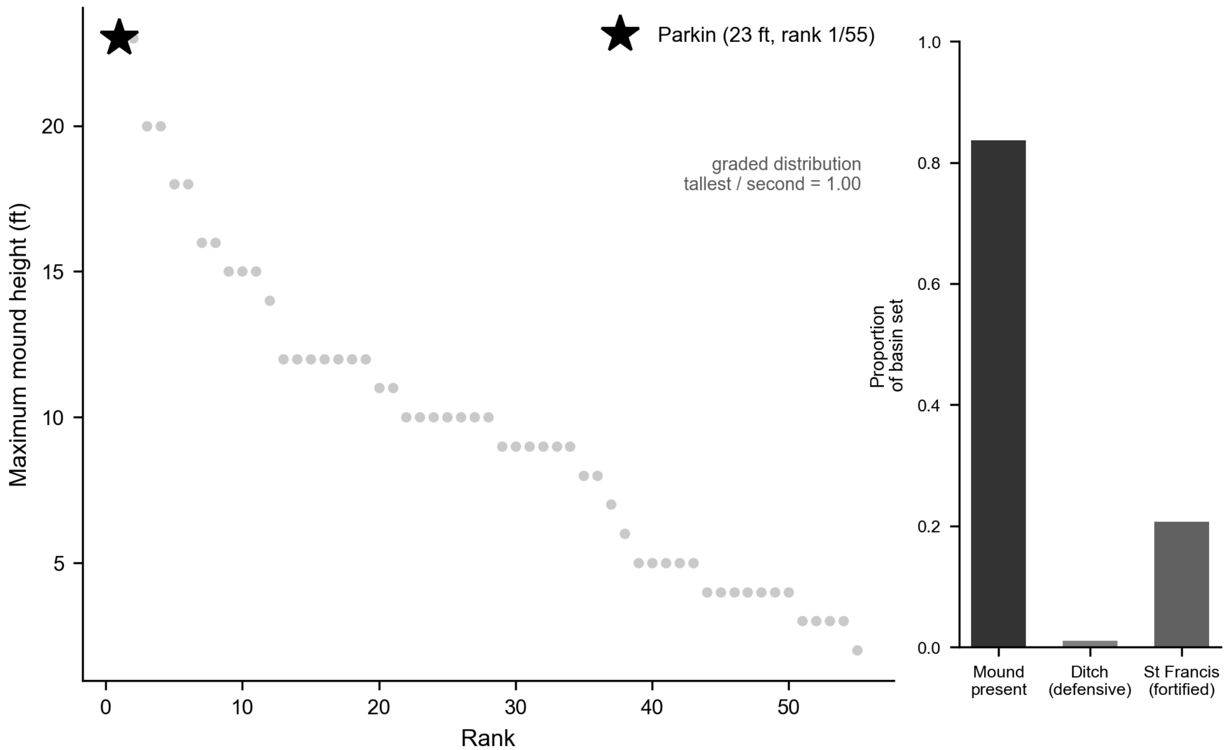

Figure 7. Settlement cross-check. Main panel: within-basin ranking of maximum mound height, the recorded, like-for-like field (Parkin's mound area is under-coded as zero and is not used). Parkin has the tallest mound (23 ft, rank 1 of 55). The height distribution for the mounds is graded, with no gap separating it from the next centers (tallest-to-second ratio 1.0), indicating that Parkin is a prominent center rather than a redistributive apex. Inset: the proportion of sites with a mound, a defensive ditch, and St-Francis-type fortification across the basin set (n = 92).

The non-ceramic evidence converges on the same conclusion. Mound size within the basin is graded rather than redistributive, and Parkin's dominance is one of scale rather than of an administrative tier set apart from the rest. The production and exchange of Mill Creek and Dover chert hoes, the capacity-linked tools whose control would mark an economic hierarchy, was decentralized at the regional scale, with distribution following a down-the-line gradient rather than redistribution from a central place (Cobb 2000). This is a regional pattern in which the St. Francis basin participates rather than a basin-internal measurement, so we treat it as consistent with, not independent proof of, the absence of economic centralization at Parkin. A separate nonmetric ordination of overlapping late-period assemblages finds that the named phases form a gradational continuum rather than discrete, bounded clusters, the cross-method counterpart of our seriation fragmentation (Mainfort 2003). These lines point in the same direction. Two of them, the settlement organization and the lithic exchange, are independent of the ceramic data, while the ordination shares that substrate and corroborates the seriation result by a different method rather than independently.

Discussion

The three lines of evidence converge on a single conclusion: the Parkin phase is not a bounded interaction group, and Parkin is not the capital of a polity. Within the St. Francis basin, Parkin is the largest and most monumentally elaborated center. Yet, its decorated-ceramic record shows no closure of interaction into bounded groups, and the structure present is reproduced by neutral drift across the basin's geography. In contrast, a bounded-groups process would have left a far sharper mark. The settlement and exchange records say the same: mound size is graded rather than redistributive, and the production of capacity-linked chert hoes was decentralized, with their exchange following a down-the-line gradient rather than central redistribution (Cobb 2000). We are explicit about the scope of the ceramic claim. The transmission analysis bears on whether interaction closes into bounded, like-with-like groups, not on group-level fitness differences or a new unit of selection (Richerson et al. 2016), which assemblage data cannot address. It likewise does not isolate demographic structure, which can shape assemblage diversity independently of any closure of interaction (Henrich 2004; Powell et al. 2009). The phase is built on a single observation: assemblages resemble their near neighbors more than their distant ones. That pattern is exactly what spatially structured drift produces on its own. At the scale at which the phase is drawn, it therefore marks continuous interaction shaped by geography rather than a bounded society. Parkin's ceremonial and demographic prominence was real. The polity that prominence is taken to imply is not in the transmission record. The decoupling has an independent precedent in the region's own ethnohistory. The best-documented Mississippian paramountcy, the sixteenth-century province of Coosa, spanned two ceramic traditions and site clusters that were separable at the phase level, so what bound them together was political affiliation rather than shared decorated style (Anderson 1996:250–251). A known polity need not coincide with a single ceramic phase, and the converse inference, from one phase to one polity, carries no warrant on its own. This interpretation is not foreign to the Central Valley literature either. In the standard regional synthesis, McNutt (1996:250) characterized the distinctions among the late Mississippian phases as little more than local elaborations, ceramic drift, on a single shared pottery complex.

That characterization can be made constructive rather than merely consistent with the data. Run the finite-population network-drift model forward on the real configuration of the assemblages, with copying that decays with along-waterway distance and no boundary imposed, and the structure that the phase concept interprets as a society reappears on its own (Lipo et al. 2021). Across the 55 decorated assemblages of the wider lower Mississippi Valley, the model repeatedly produces a few spatially coherent, phase-like groups (Figure 8). The deposits are not contemporaneous, which is why they can be seriated at all and why some carry more of the late classes than others, so each synthetic assemblage is drawn time-transgressively at the run-time that matches its seriation position. Holding time fixed to isolate the spatial signal, drift yields two to five groups whose number grows with geographic extent; restoring the staggered deposition ages collapses the count toward two time-defined groups, because assemblages caught at different points in the sequence carry different repertoires. Either way, the number of drift-generated groups falls well short of the seven named phases the regional scheme draws across the same assemblages. The between-group cultural F_ST of those groups settles at the observed level rather than the higher value a bounded group would leave. The named phases, therefore, subdivide the valley more finely than the transmission structure supports. Where culture history draws seven bounded societies, drift on the geography yields at most a handful of coherent groupings, so most of the phase boundaries mark divisions that a continuous interaction field does not carry. Whether any single named phase exceeds this drift expectation is a question we can now pose one phase at a time. That spatial and population structure, not population size, governs how drift partitions diversity in finite structured populations is established in both the cultural-transmission and population-genetics literatures (Barton 1996; Premo 2016; Schneider et al. 2016; Wright 1943). A localized neutral process is readily mistaken for biased transmission when diversity is taken at face value (Premo and Scholnick 2011). Structure shapes diversity under selective dynamics as well, through a different mechanism of maintenance and recombination (Derex et al. 2018; Derex and Boyd 2016; Saxer et al. 2009). The result is not contingent on a tuned parameter set. Across 432 runs spanning a grid of interaction length, between-node mixing, and innovation rate, two or more spatially coherent groups appear in every run, and the between-group F_ST sits at or below the observed value in 85 percent of them. The St. Francis basin that anchors this paper behaves the same way as a positive control (Figures S7 and S8).

The same model lets the named phases be tested one at a time, in the culture historian's own terms. Taking Parkin as the focal case, membership in its group is a probability rather than a fact: across the 500 runs, each assemblage shares Parkin's drift-detected group some fraction of the time, a graded field that falls off smoothly with river distance from Parkin rather than the bounded set a phase map implies (Figure 9A). The Parkin-phase assemblages do separate from the others by this measure. That separation is what drift predicts, because they lie near Parkin and so co-occur with it more than distant assemblages do (Figure 9B). The discriminating test is whether the between-group cultural F_ST across the Parkin-versus-others partition exceeds the drift null. It sits at the extreme upper tail of the null and survives most single-site deletions. The contrast is nonetheless small: the null's own tail reaches past it, and time-averaging, which deflates between-group variance, is not modeled in the null (Figure 9C, 9D). It is therefore at most a weak hint of structure beyond drift, not a demonstrated social boundary. Geography and time account entirely for the pattern, and the apparent groups are products of a single neutral process.

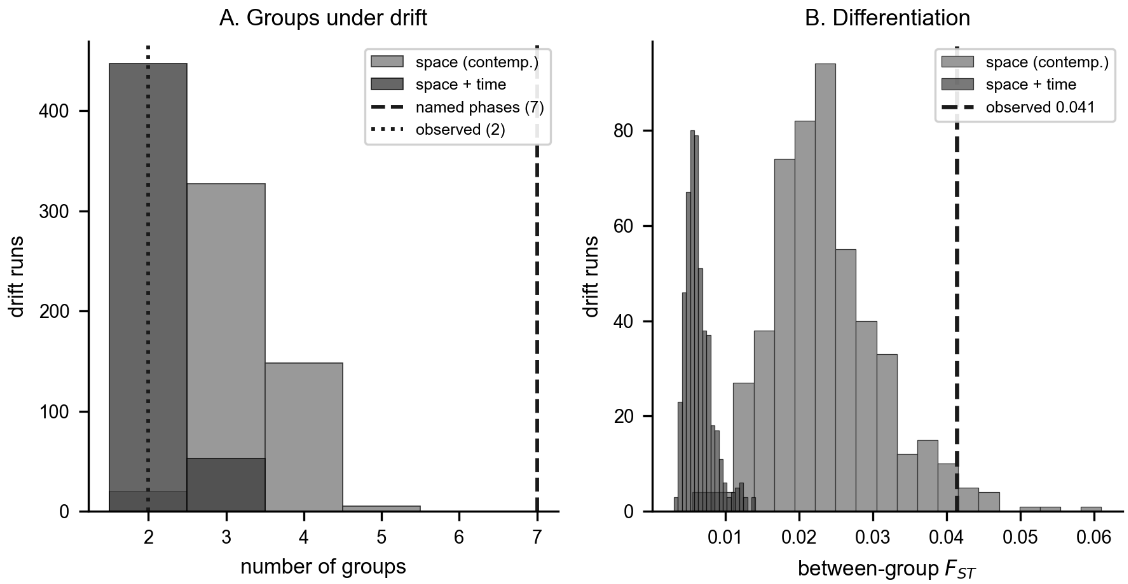

Figure 8. Phase-like groups produced by neutral drift across the wider lower Mississippi Valley, with no bounded group imposed, summarized over 500 realizations. The finite-population network-drift model runs forward on the river-network geography of all 55 Mainfort-PFG decorated assemblages, and groups are detected as greedy-modularity communities on the Brainerd-Robinson similarity graph (Supplemental Text). In the model, each site copies decorated classes primarily from itself and, at a low rate, from other sites, with a probability that decays with river distance, while innovation turns the repertoire over through time; synthetic assemblages are sampled time-transgressively at each site's seriation position and subsampled to match the observed counts. (A) Distribution of the number of groups across the 500 runs, under contemporaneous (pure-space) and time-transgressive sampling, with the seven named culture-historical phases and the observed cluster count marked. Pure spatial drift yields two to five groups, and the count grows with geographic extent; time-transgressive sampling collapses it toward two. Either way, drift produces far fewer groups than the seven named phases, so the phases subdivide the valley more finely than the transmission structure supports. (B) Between-group cultural F_ST across runs against the observed level (0.04); the observed differentiation sits at the upper edge of the contemporaneous-drift distribution, so drift reproduces it. Parameters are the positive-control corner (interaction length 24 river-km, mixing 0.02, innovation 0.012); the basin positive control and the parameter-grid robustness are in Figures S7 and S8.

This finding provides the strongest case for the alternative hypothesis. The most direct argument for an emerging hierarchy holds that Parkin sits on agriculturally poor soil and so must have drawn tribute from the productive villages around it (Phyllis A. Morse 1990). The inference is specific and deserves credit for that, but it is circular. The tribute it concludes is the tribute it assumes, and no independent evidence of tribute flow accompanies it. Parkin's position at the confluence of the St. Francis and Tyronza is also explained by control of movement and communication, as well as by extraction. The transmission and exchange records both show the absence of the centralized control the tribute model requires. A second argument points to skeletal evidence from the lower Mississippi (Yazoo) Delta rather than the St. Francis basin, indicating a better diet at a mound center than at a nearby satellite village, consistent with differential access within a hierarchy (Ross-Stallings 2017). The contrast is real but thin: the satellite sample numbers fourteen individuals, the result is presented as tentative, and a health difference between two sites does not require an extractive hierarchy to produce it. Neither argument supplies the regional, structural signature that the drift-versus-groups comparison looks for and does not find.

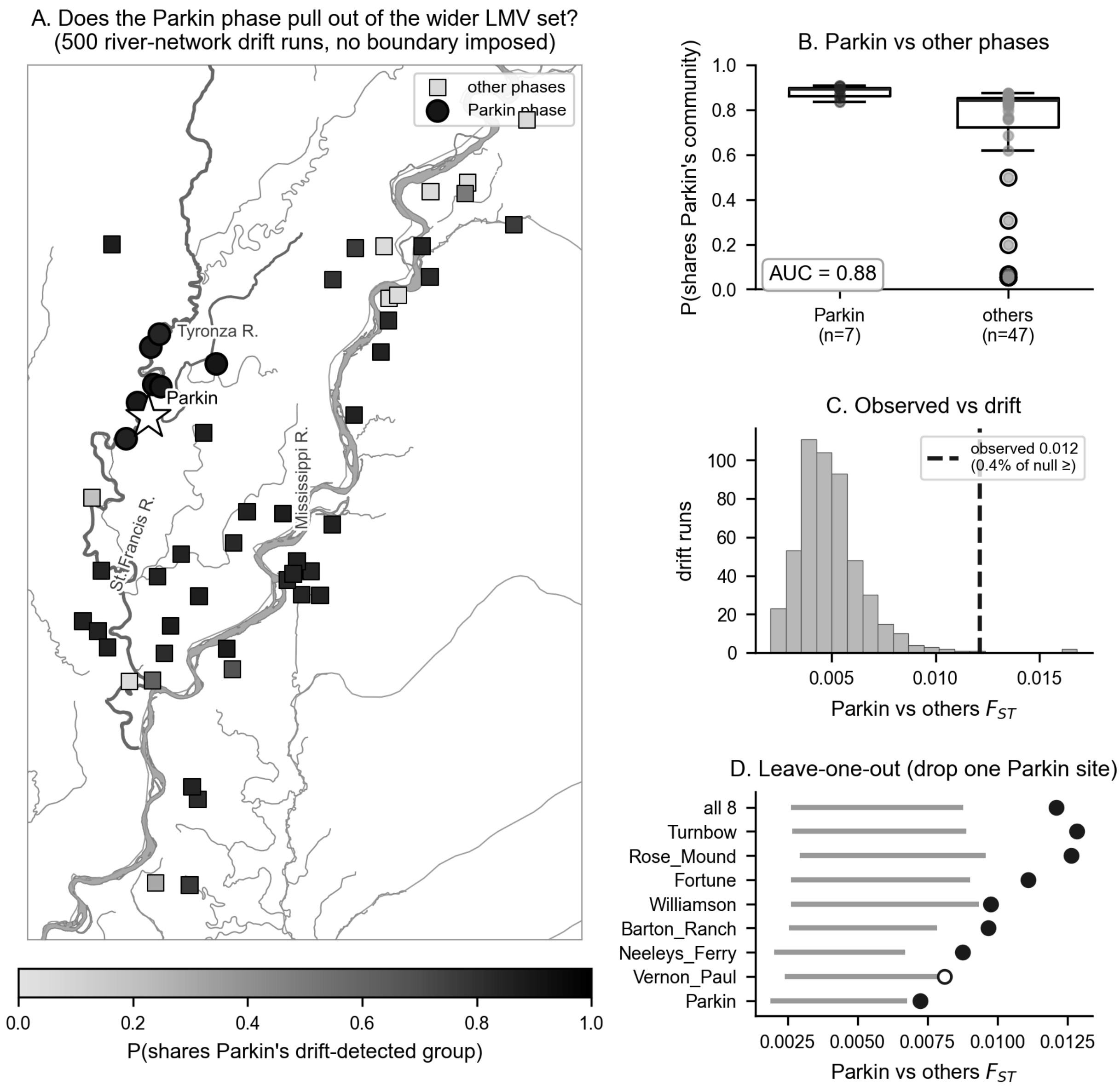

Figure 9. Does the Parkin phase separate from the other named phases by more than drift? All 55 Mainfort-PFG decorated assemblages, labeled by culture-historical phase after Mainfort (1996b), under time-transgressive river-network drift with no boundary imposed (500 realizations). Because group labels differ between runs, Parkin's group is summarized as the fraction of runs in which each assemblage falls in the same drift-detected group as Parkin. (A) Each assemblage is shaded by the probability that it shares Parkin's drift-detected group (star), with Parkin-phase assemblages drawn as circles and other phases as squares; the probability is a graded field that falls with river distance from Parkin. (B) That co-membership for the Parkin phase against the other phases (AUC 0.88) reflects the separation drift predicted by geography. (C) Observed Parkin-versus-others between-group cultural F_ST (dashed) against the drift null; 0.4 percent of runs reach or exceed it, and the null's tail extends past it. (D) Leave-one-out jackknife: the observed F_ST with each Parkin assemblage relabeled into the comparison group, against the 2.5 to 97.5 percentile band of its matched per-drop null. Open points fall inside the band (drift-consistent), filled points outside; seven of the eight single drops stay outside, so the small contrast does not rest on one assemblage.

Three limits bound the result, and the recovery experiment forces the most important of them to the front. At this record's resolution, the discrimination between a bounded group and drift rests on a single signature, cultural F_ST. That signature is confounded by a sample-size trend along the seriation axis (Spearman rho = +0.59). We removed the confound by subsampling every assemblage to a common count, which collapses the raw F_ST trend from +0.60 to +0.01. We calibrated the test on synthetic data matched to the record, in which F_ST recovers a moderate-strength closure (s of about 0.5, rising to 0.7 once the record's time-averaging is imposed), while maintaining a false-positive rate of 5%. The size-controlled empirical F_ST is uninformative, its bootstrap interval spanning no closure to a moderate signal. The negative is therefore informative for a closure of moderate or greater strength, which would have registered as a resolved positive, but does not. It cannot exclude a weaker closure than a record of 29 time-averaged assemblages, which is underpowered to resolve. We state the claim at that bound rather than beyond it. The other three signatures do not track closure reliably here. The neutral departure fails because it is non-monotonic in conformity and sensitive to sample size. The spatial boundary fails because the data support too few spatial clusters, and seriation coherence responds only weakly. We report them as corroboration rather than as independent confirmations. This reduction is a property of the record. When a known closure signal is injected instead through a within-group conformity pathway or a non-spatially-organized social-fragmentation pathway, neither the neutral nor the seriation signature recovers it either, so the three weak signatures fail at this resolution regardless of how closure is produced (Supplemental Text). The price is that the negative bounds are only closure expressed as between-group divergence. The closure carried purely by conformity, or by social structure not aligned with the spatial clusters, would lie below detection in this record at any strength. The second limit is the ordinal chronology. The basin radiocarbon corpus pools to five seriation anchors across a calibration plateau, so the axis is a relative order oriented toward later time at Spearman +0.70, imperfectly. The discriminating quantity, the size-controlled F_ST trend, is a magnitude that does not depend on the axis orientation. The settlement and exchange lines do not depend on the ceramic chronology at all. The third limit is time-averaging, which damps between-assemblage variation and reduces apparent turnover (Coco and Iovita 2020; Kidwell and Tomašových 2013; Perreault 2018; Tomašových and Kidwell 2009, 2010; Youngblood et al. 2023). Rather than argue that this makes the absence conservative, we built the damping into the recovery experiment directly, where it raises the detection threshold from s of 0.5 to 0.7, the quantified cost of the record's temporal mixing. The verdict is stable across the watershed (19 assemblages) and corridor (29) basin definitions. The central methodological finding is that telling a bounded group from drift must be calibrated to the record to which it is applied. The same four signatures that discriminate cleanly on idealized data reduce, on a real time-averaged record of this size, to a single confound-controlled signature. Saying so is what licenses interpreting the structure as drift rather than weakening the claim.

The contribution has two parts that should be kept distinct. The four signatures distinguish a bounded interaction group from drift in principle, and an idealized simulation confirms that closure drives them together while drift and the other mimics do not (Figure S4). The calibration is the part that travels to real data: it asks, before any similarity structure is interpreted as social, which signatures a given record can support and how strong a closure each could detect. The Parkin record shows the test reducing to its single most robust component, cultural F_ST. We are explicit about what that costs. Had F_ST shown a positive trend, it alone could not have separated a real group from environmental patchiness, which also drives between-group variance (Table 1). In the positive case, the generative drift-versus-groups comparison, not F_ST alone, would have had to carry the discrimination. The negative is cleaner because the present structure is positively matched by drift and overshot by bounded groups. This result shows that social structure may be measurable through decorated ceramic assemblages. We can also assess whether the Parkin phase was a polity by examining whether interaction had closed into a bounded group to a degree that the record can resolve. For the Parkin phase, the answer is that it had not, within the bounds supported by the record. The polity the De Soto encountered seems to reflect distinctions drawn in ceremonial practices, fortification, and settlement size, rather than one written into the structure of ceramic transmission.

One result resists the drift interpretation, if only faintly, and deserves to be named rather than buried. The single place in the valley that carries a faint signal of structure beyond drift is the Parkin phase, and Parkin is the one monumental node, the largest mound, the palisade, the confluence position, the candidate Casqui. A center beginning to differentiate as a distinct node, rather than a finished, bounded group, would leave just such a signal. This is what the origination stage of the two-regime model predicts: a local return generates a group-beneficial pattern at a node before assortment becomes binding and propagates into higher-level structure, with contact and the protohistoric depopulation truncating the sequence before it could consolidate. We do not claim this happened. The contrast is small, the drift null reaches past it, and the time-averaged surface record of eight Parkin-phase assemblages cannot support the claim. The possibility that Parkin was an incipient node nonetheless warrants direct investigation with data classes the present record lacks: stratified, excavated assemblages with independent radiocarbon, to test whether the differentiation grows through time rather than holding flat; compositional and petrographic sourcing of the ceramics, to test whether production and exchange were localizing on the center; and the non-ceramic lines, the chert-hoe and shell exchange networks (Cobb 2000) and the mortuary and dietary record (Ross-Stallings 2017), to show whether economic and social differentiation track the ceramic hint or run independent of it. Whether Parkin was beginning to pull away from its neighbors, or sits where drift on the river network happens to concentrate interaction, is a question this record raises but cannot settle.

The test also extends beyond the basin (Figure 10). Applied to the southeast-Missouri assemblages compiled by Williams (1954), the within-region decorated-ceramic structure is again drift-consistent once the earlier Woodland component is set aside. Therefore, the absence of bounded groups is not peculiar to the St. Francis basin (Figure 11). In both regions, the observed between-cluster F_ST sits at the neutral-drift level and far below the bounded-groups expectation (LMV +0.02, CMV +0.03, against bounded-groups nulls of +0.11 and +0.08). The assemblages intermix rather than separating into clusters. Between the two regions, a genuine discontinuity appears across the roughly 50 km gap separating the central-valley and lower-valley assemblage clusters, with decorated similarity declining by far more than neutral spatial drift would predict for the same geometry. The two regions carry mutually exclusive, distinctive decorated repertoires (Figure S6). This is the scale-dependent pattern the test is designed to detect, with no closure within regions and a sharp difference between them. We report it as a pattern rather than a verdict, because it is confounded with time.

The distinctive central-valley classes are early- to middle-Mississippian, and the distinctive lower-valley classes are late (Phillips 1970; Williams 1954). The step at the boundary, therefore, cannot be separated from the early-to-late turnover of the repertoire, as reflected in time-averaged surface assemblages. Resolving whether the inter-regional boundary is synchronic and social or a chronological succession would require independent radiocarbon control of the central-valley assemblages, which the present compilation lacks. A second confound compounds the first. The lower-valley assemblages were collected and classified by Ford, later by Phillips, Ford, and Griffin.

In contrast, the central-valley assemblages were collected and classified separately by Williams, so the two blocks differ in analyst, sorting convention, and class definition as well as in space and time. The shared-class restriction guards against the crudest version of this artifact: a boundary built only from classes unique to each typology would vanish when only the classes both traditions record are compared. The boundary does weaken under that restriction without disappearing. Even the shared classes, however, may have been sorted to subtly different criteria by the two traditions. We therefore cannot rule out that the apparent central-valley and lower-valley boundary is in part an artifact of two separate analytical histories, and that the possibility of continuous interaction across the whole region, with no real boundary anywhere, remains open. That possibility is part of the question this study raises, rather than a nuisance to be set aside, because a boundary in the data is not yet a boundary in the past. The paper's point at the local scale applies with equal force between the regions.

Beyond the Parkin case, the test proposes a standard of measurement for an evolutionary archaeology that has long been clearer about what evolves than about how to measure it (Dunnell 1995a). A signature that interaction has closed into a bounded group should be theoretically sufficient, dynamically sufficient, and empirically sufficient, the last meaning matched to the quality of the record rather than to an imagined snapshot. Ours meet the first by construction, each entailed by the transmission model rather than borrowed by analogy (Lipo et al. 2015; Neiman 1995). The third is that we test rather than assume, which is the point of the recovery experiment: empirical sufficiency is a property a signature has only if it can register a signal at the resolution preserved by the deposit. On the Parkin record, only between-group variance qualifies (Perreault 2019). They meet the second only in part, because the signatures measure states along

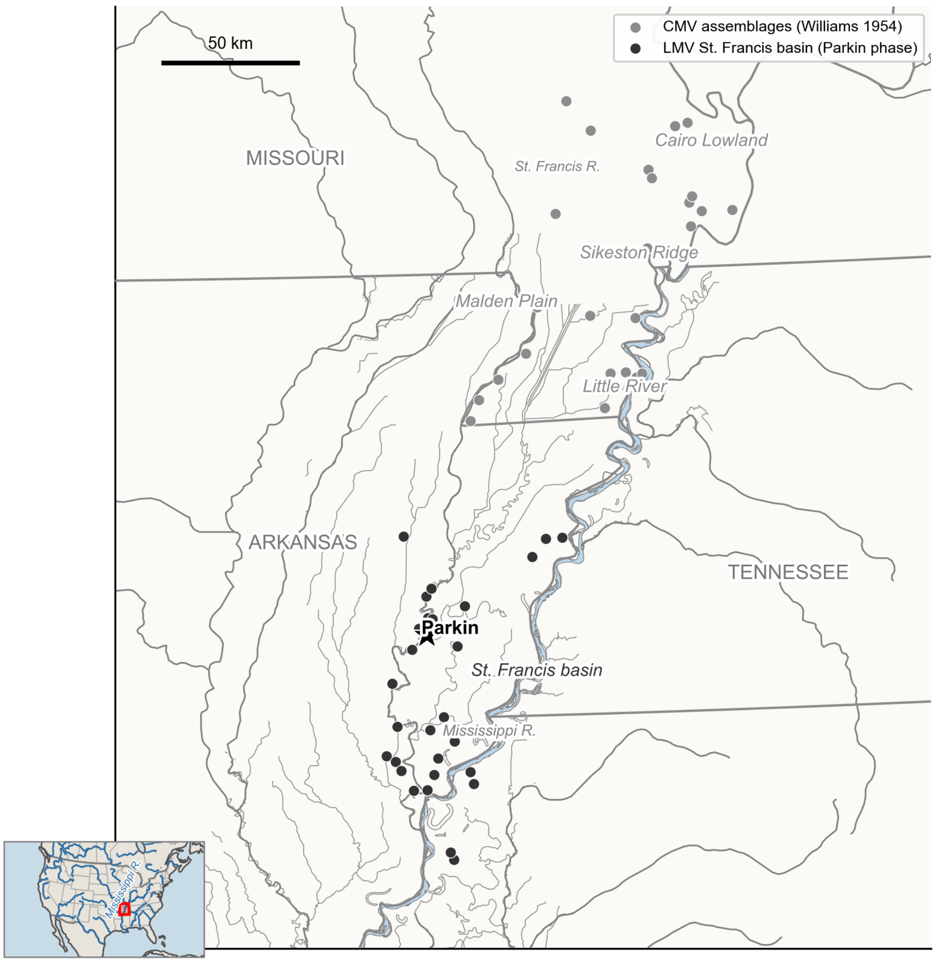

Figure 10. Regional setting of the two-scale comparison. Decorated-ceramic assemblages of the central Mississippi Valley: the southeast-Missouri survey of Williams (CMV, blue) and the lower-valley St. Francis basin Parkin-phase assemblages (LMV, orange), with the gap between the two clusters of assemblages and Parkin starred. Labels mark the named physiographic phase regions (Cairo Lowland, Sikeston Ridge, Malden Plain, and Little River in the central valley, with the St. Francis basin in the lower valley). Hydrography combines the Lower Mississippi Survey layer with Natural Earth, state boundaries, and the North America locator inset, which are from Natural Earth. The central-valley assemblages were collected and classified by Williams, and the lower-valley assemblages in the Phillips-Ford-Griffin tradition, the separate analytical histories discussed in the text.

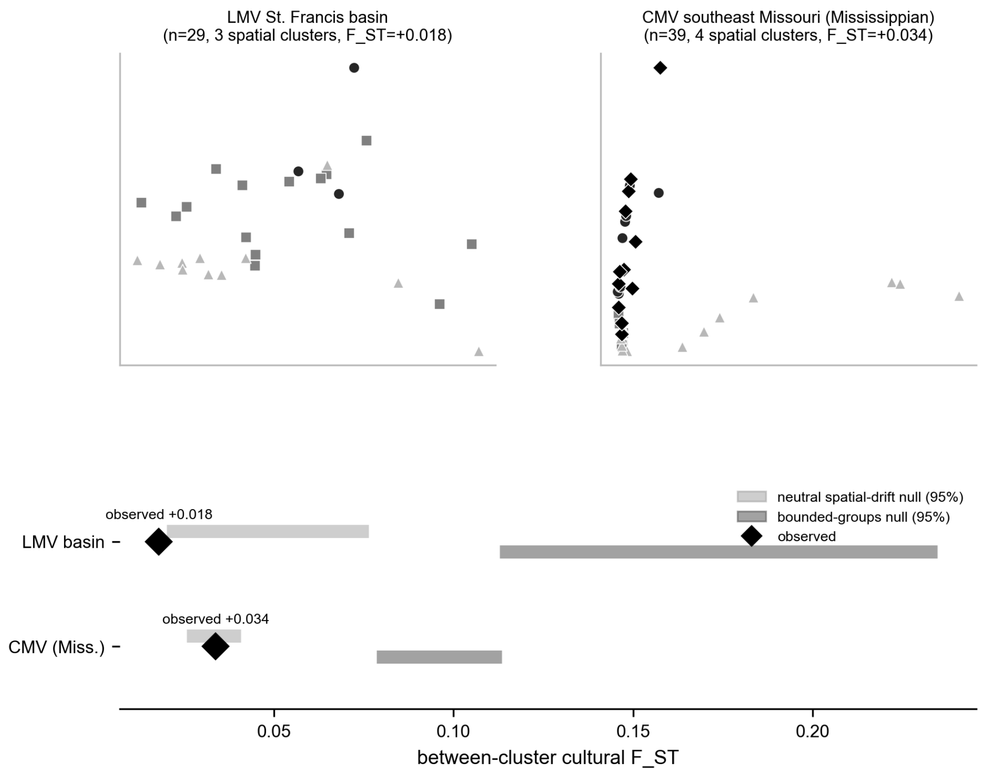

Figure 11. Within-region interaction structure: neither region forms bounded groups. Top: multidimensional scaling of decorated Brainerd-Robinson dissimilarity for the LMV St. Francis basin and for the CMV southeast-Missouri assemblages (Mississippian classes only, so the comparison is not driven by the Woodland-to-Mississippian gradient), points colored by within-region spatial cluster, and the assemblages intermix rather than separating into clusters. Bottom: the observed between-cluster cultural F_ST for each region (diamond) against the 95% nulls of a neutral spatial-drift model (blue) and a bounded-groups model (orange) run on the real coordinates and subsampled to the observed counts. In both regions, the observed F_ST sits at the drift level and far below the bounded-groups expectation, so the within-region structure is drift-consistent at both scales.

the seriation sequence rather than the transformation between them. We therefore add two time-domain tests that jointly satisfy dynamic and empirical sufficiency. A generative, time-averaging-aware inference fits a frequency-dependent copying model to the whole sequence, with the record's time-averaging built into the model. It recovers a transmission bias tightly constrained near neutral (95 percent interval -0.02 to +0.06, including zero), which rules out the conformist dynamic a marker would require (Crema et al. 2016). A tempo-and-mode comparison finds the aggregate diversity trajectory best described by neutral drift rather than by a directional, stabilizing, or mean-reverting alternative (Hunt 2006; Hunt et al. 2008). Neither recovers a transition that the static criterion missed, so the negative now holds in the time domain as well as on the static states. A third probe, for the early-warning signature of a critical transition, is underpowered at this ordinal resolution and does not meet the empirical sufficiency criterion, so we report it in the supplementary code but do not rely on it (Dakos et al. 2012; Scheffer et al. 2009). The detail is in the Supplemental Text.

Conclusion

The phase is the unit on which central-valley culture history is built, and it carries a silent claim that assemblages grouped by seriation and spatial proximity record a bounded society. For the Parkin phase, that claim does not survive measurement. Parkin is the basin's largest and tallest-mounded center. No evidence of settlement, exchange, or ceramic transmission marks it as the apex of a polity. The greater similarity among nearby assemblages that defines the phase is reproduced by neutral drift on the basin's geography, the ordinary residue of continuous interaction structured by where sites sit. A bounded interaction group would have a sharper structure than the measurements indicate, and Parkin sits across orderings as a transmission bridge rather than at the closed core of one. Extending the test to the wider valley, a decorated-ceramic boundary that exceeds drift appears only between the central-valley and lower-valley blocks. Even there, it is entangled with the early-to-late turnover of the class repertoire. It cannot yet be interpreted as a synchronic social division. Posed in the culture historian's own terms, against the other named phases of the valley, the Parkin phase itself shows at most a weak contrast at the edge of the drift null, robust to single-site removal but small and within the null's reach (Figure 9). The phase, at the scale where it is drawn, is largely indistinguishable from spatially structured drift. That carries a consequence the discipline should weigh: a similarity cluster is not, by itself, a society, and before a phase is interpreted as a social group, the record must be shown to distinguish a real interaction boundary from drift across geography. That calibration, not the phase, provides the basis for an explanation of social structure. When applied to the late pre-contact ceramics of the Mississippi Valley, it shows continuous interaction with prominent nodes rather than a mosaic of bounded polities.

Data and Code Availability

All data and code needed to reproduce the analyses and figures reported here are openly available at https://github.com/clipo/phases. The repository includes the decorated-ceramic class counts, the georeferencing and radiocarbon data, the complete analysis procedure, and instructions for reproducing all results and figures. The code is released under the MIT License, and the data under a Creative Commons Attribution 4.0 International license.

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